深度學習
Deep Learning

Transformers
自注意力機制模型

Instructor: 林英嘉 (Ying-Jia Lin)
2025/11/05

Course GitHub

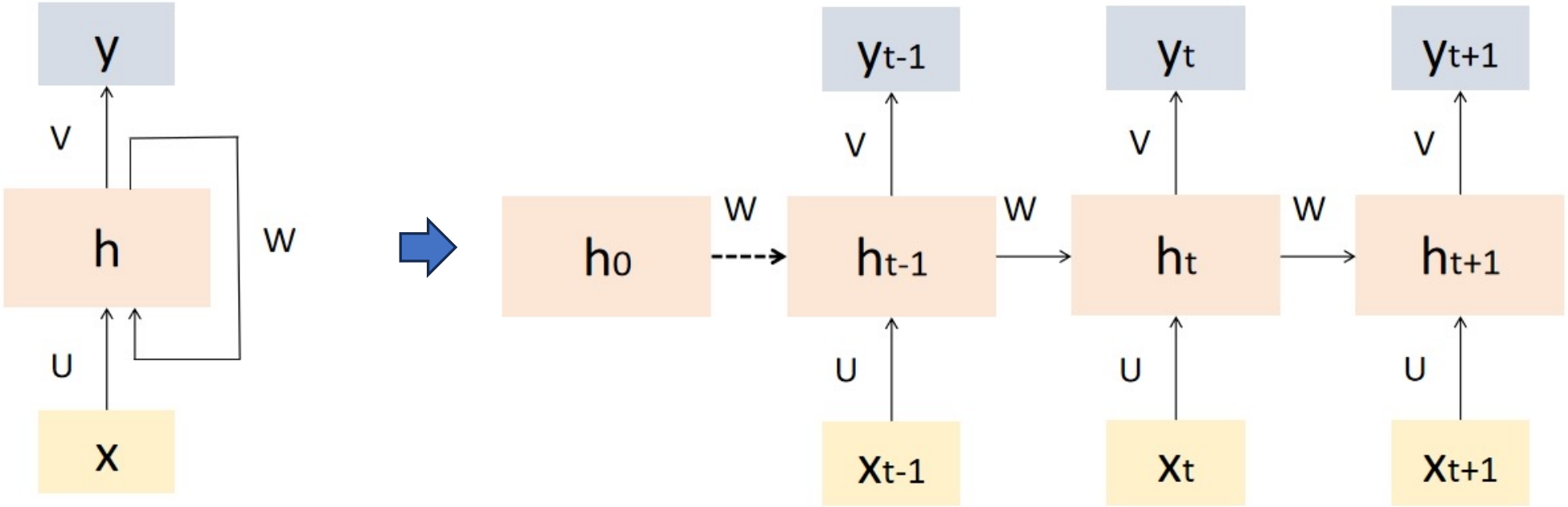
Slido # DL1105

Outline

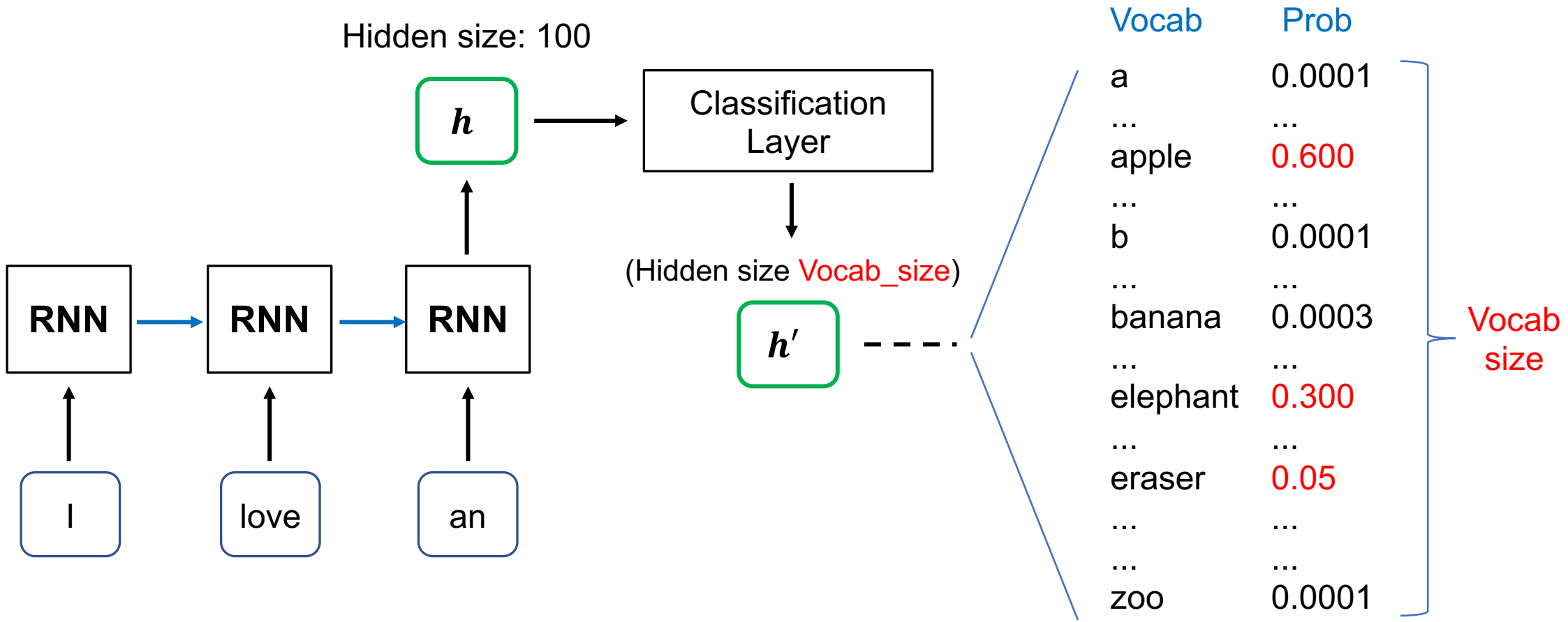
- Weakness of RNNs and CNNs [10 min]
- Transformers [90 min]
- Announcement about Term Project [10 min]
- Homework 3 [25 min]
- Quiz [15 min]

[Recap] Recurrent Neural Networks (RNN)

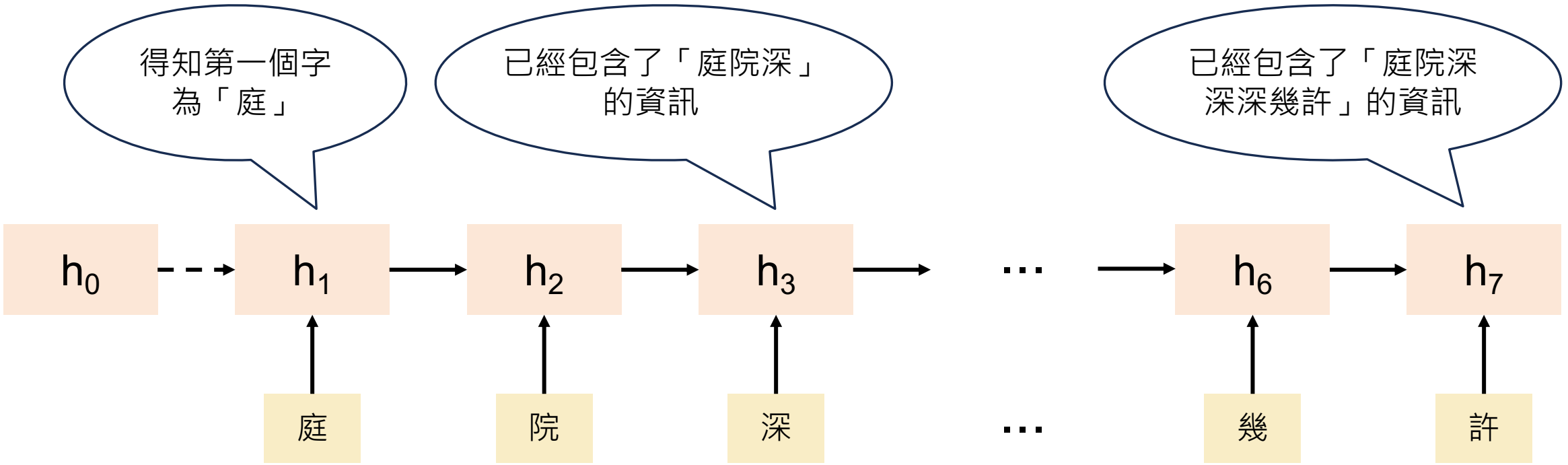
- Recurrent Neural Networks (RNNs) are a type of artificial neural network designed to handle sequential data by capturing temporal dependencies.
  - RNN 的“遞迴”是指它在每一個時間步中重複使用相同的參數（同一組權重），並把先前的隱藏狀態傳到下一個時間步



[Recap] Text Generation with RNN



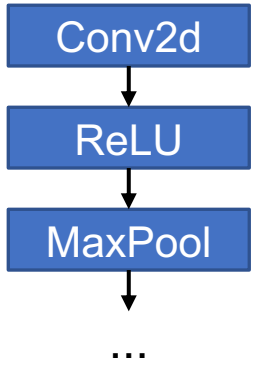
[Recap] RNN 的記憶力意義



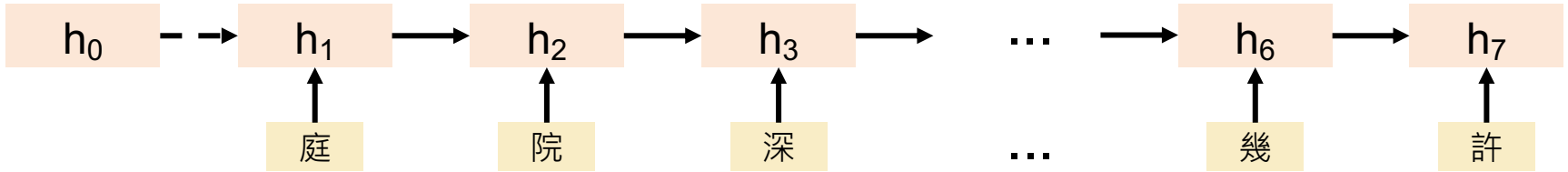
序列模型 (Sequential Model)

一層一層執行
(Recap)

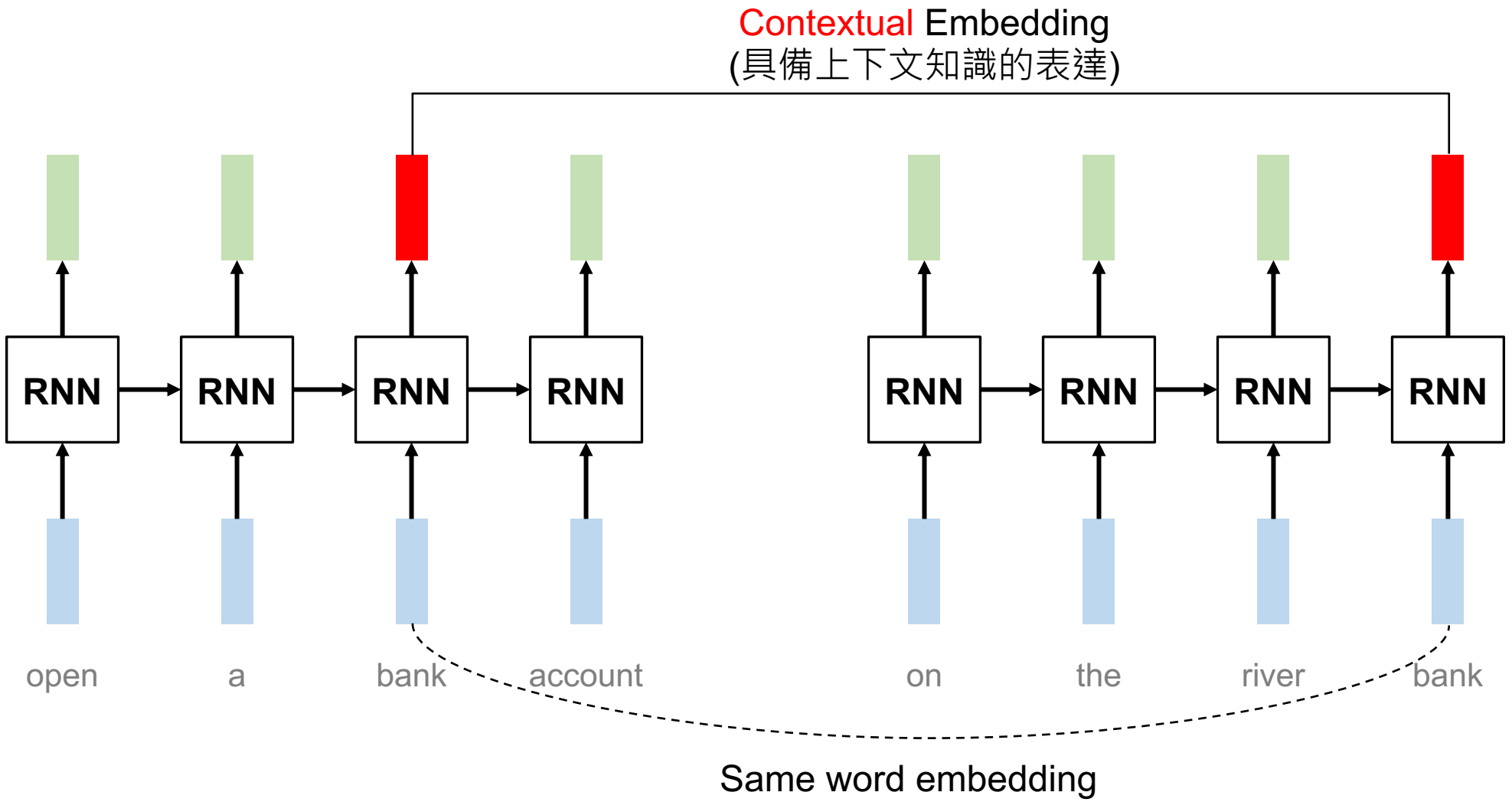
```python
class SimpleCNN(torch.nn.Module):
    """寫法2: 模型傳遞流程用 `nn.Sequential` 包起來
    """
    def __init__(self, num_classes=10, fc_hidden_size=256, dropout_ratio=0.5):
        super().__init__()

        self.feature_extractor = nn.Sequential(
            nn.Conv2d(in_channels=3, out_channels=64, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(kernel_size=2, stride=2),

            nn.Conv2d(in_channels=64, out_channels=64, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(kernel_size=2, stride=2),

            nn.Conv2d(in_channels=64, out_channels=128, kernel_size=3, padding=1),
            nn.ReLU(),
            nn.MaxPool2d(kernel_size=2, stride=2),
        )
```



序列模型：
一步一步執行，
有時間點關係

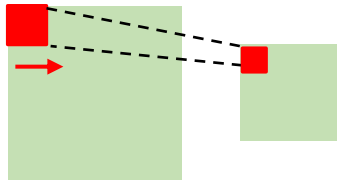


為什麼我們需要序列模型?

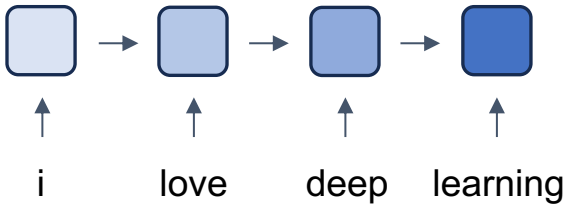


深度學習常見模型架構

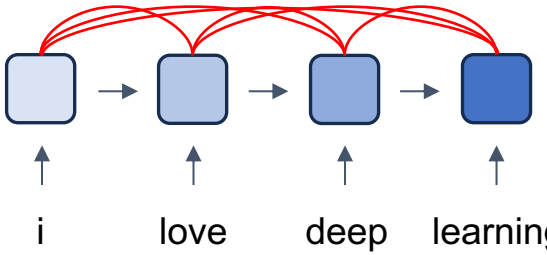
CNN
(著重局部資訊)



RNN
(著重記憶力)

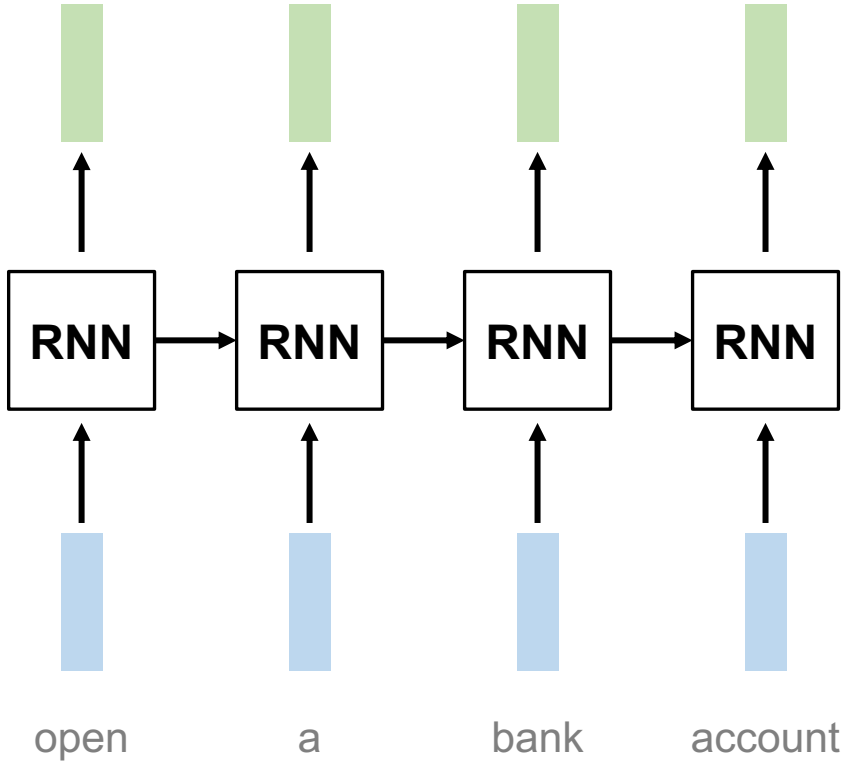


Transformer
(著重全局資訊與記憶力)

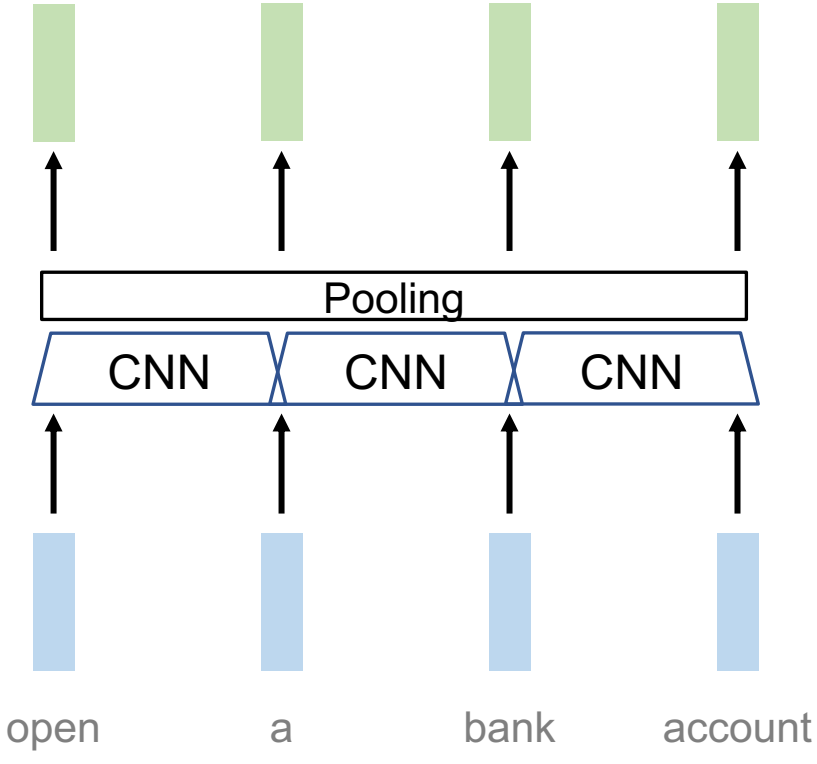


RNN 與 CNN 作為序列模型的問題

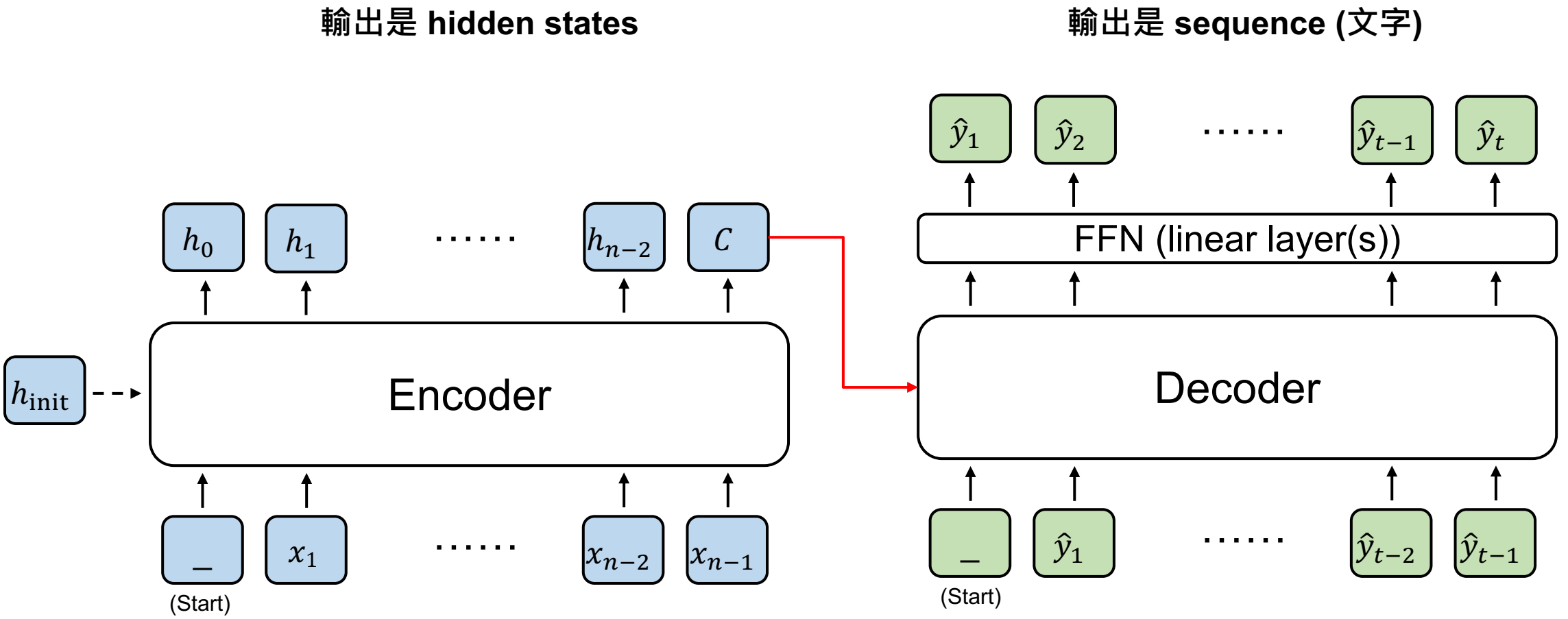
問題：RNN 難以平行化



問題：CNN 著重局部資訊



Encoder and Decoder

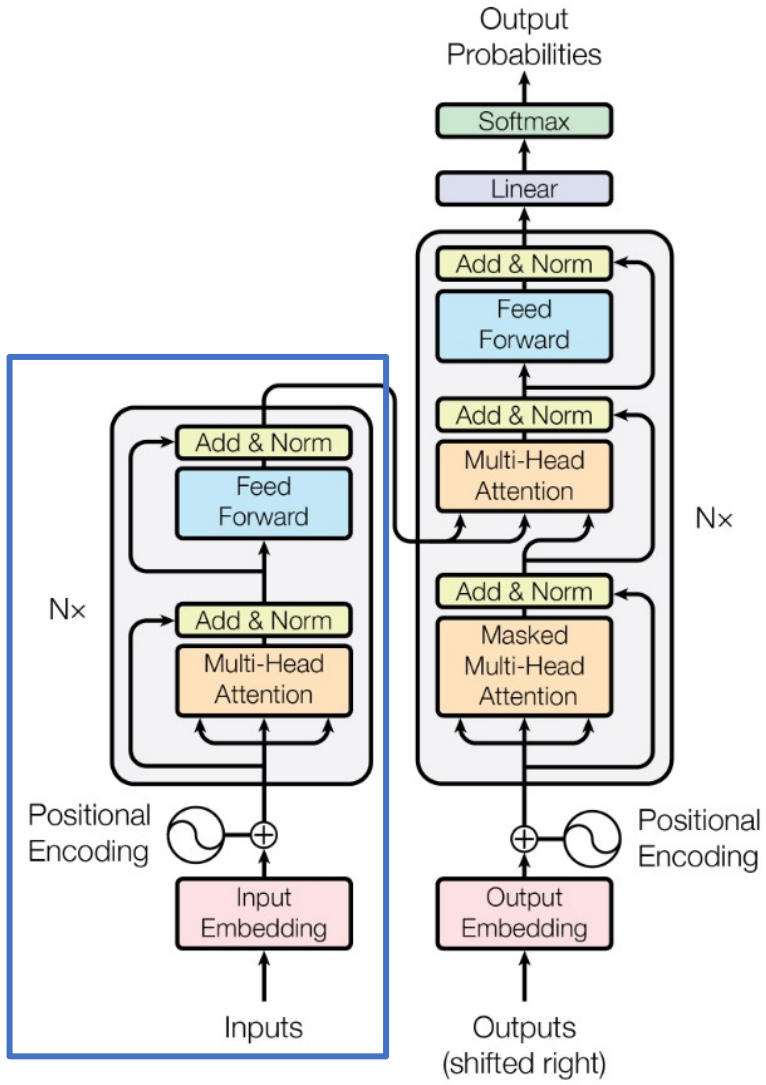


(Encoder 的輸出有很多種
被Decoder運用的方式)

Transformer

Overview of Transformer

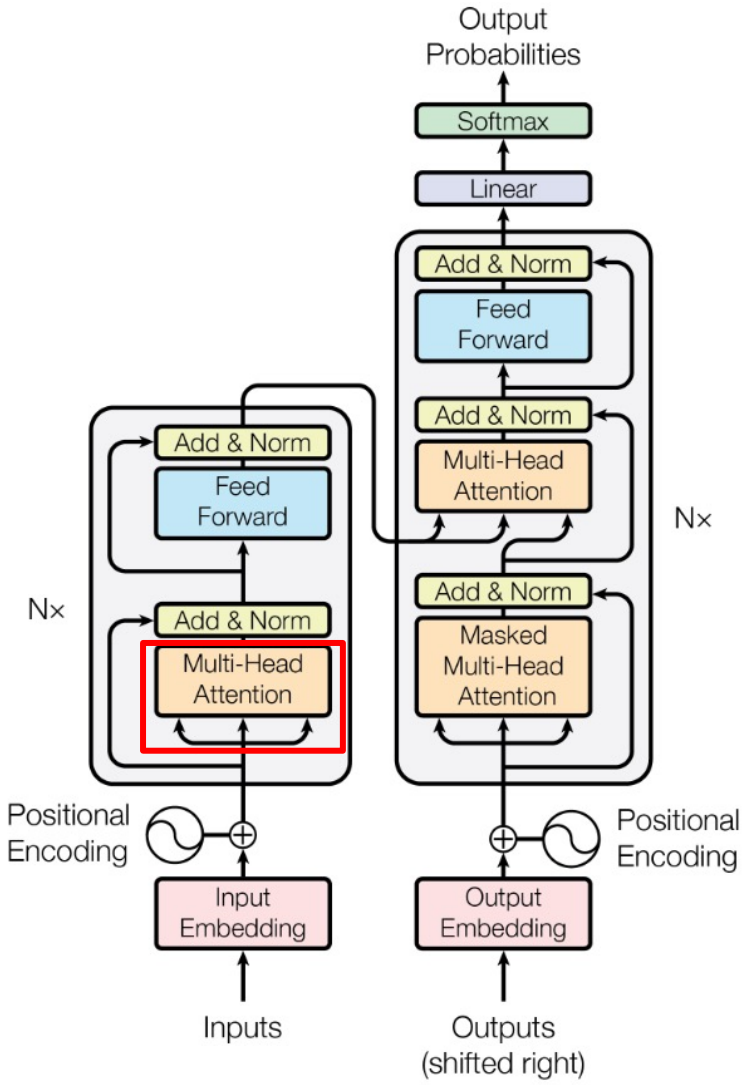
Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).



Encoder

Decoder

Outline of Transformer



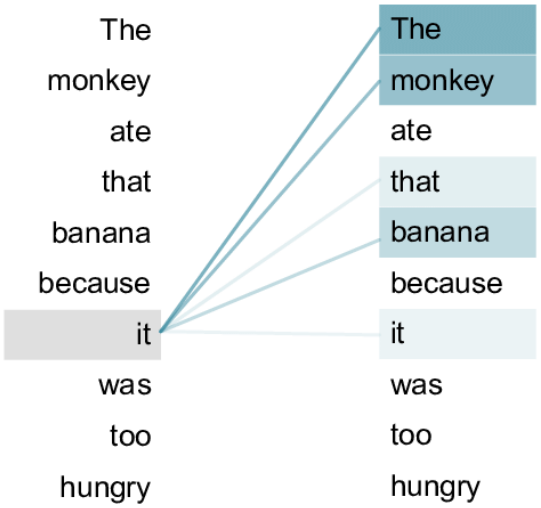
Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

Transformer

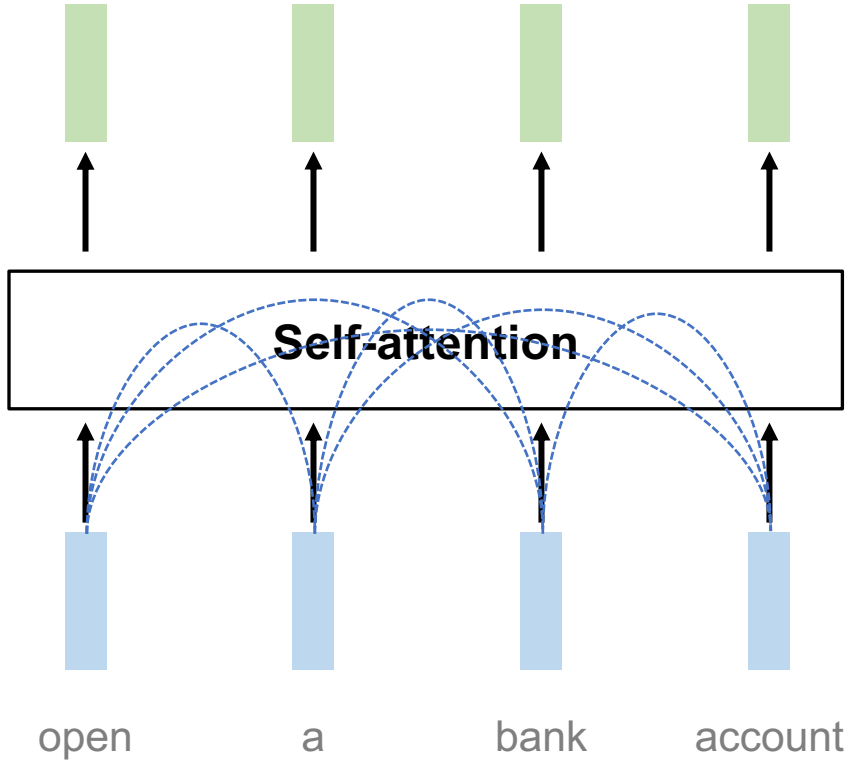
Source: An example of the self-attention mechanism following long-distance... | Download Scientific Diagram (researchgate.net)

- Transformer 來自論文：Attention Is All You Need (Vaswani et al., 2017)
- Transformer 內部的主要機制為 Self-attention

Self-attention: 句子內的每個 token 自己跟自己做 attention

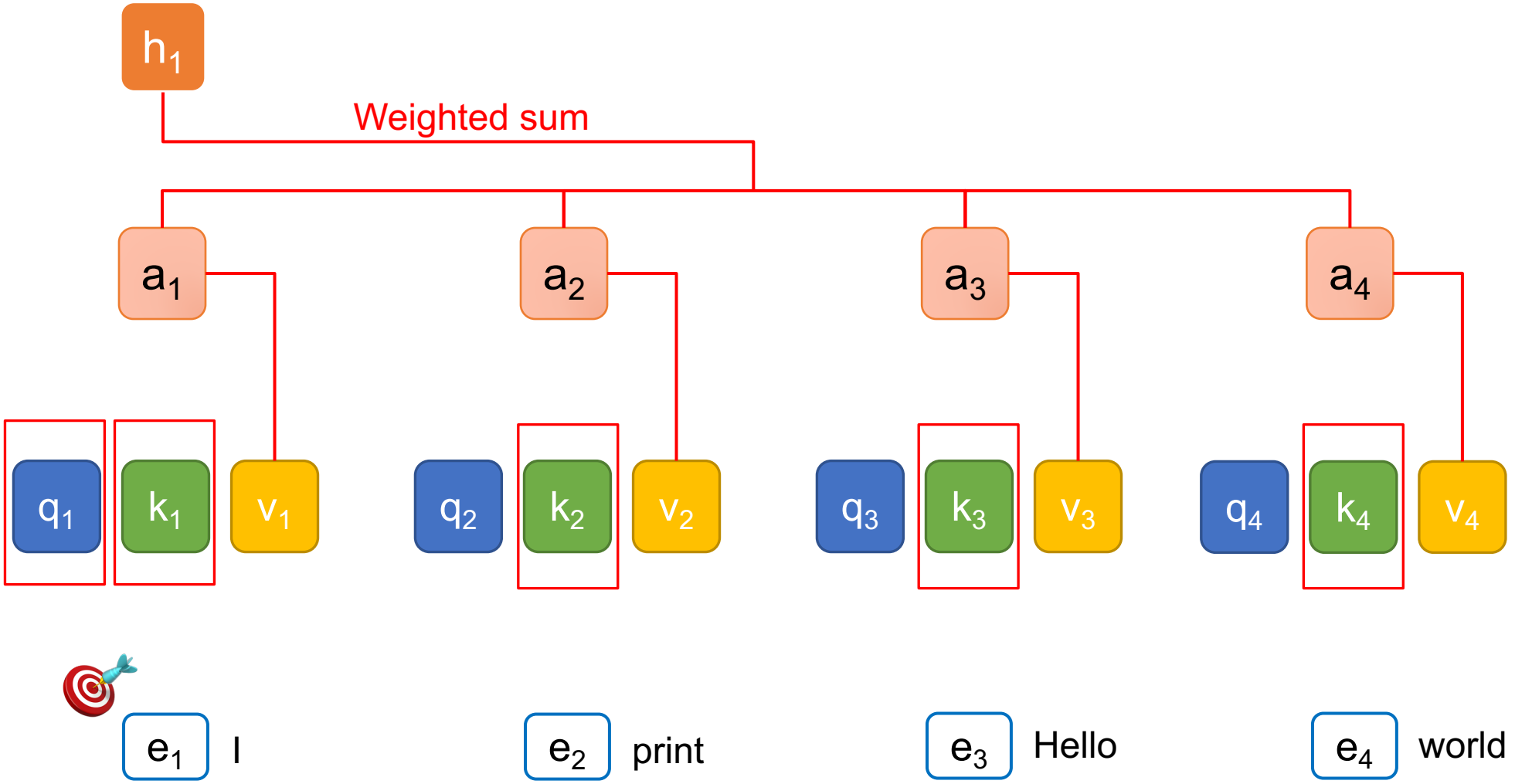


attention 進行的過程需要主動與被動



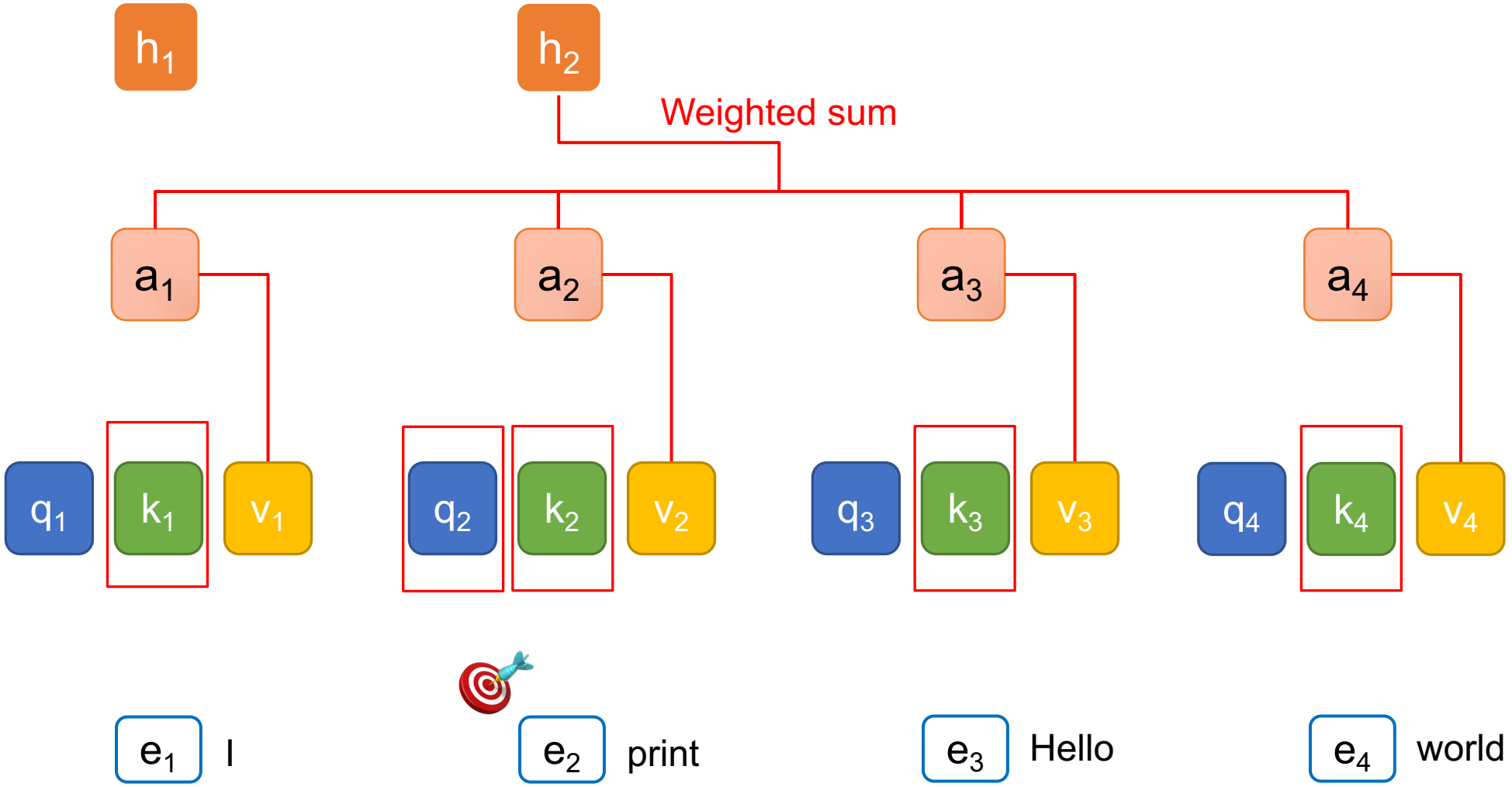
Query (Q), Key (K), and Value (V)

t = 1



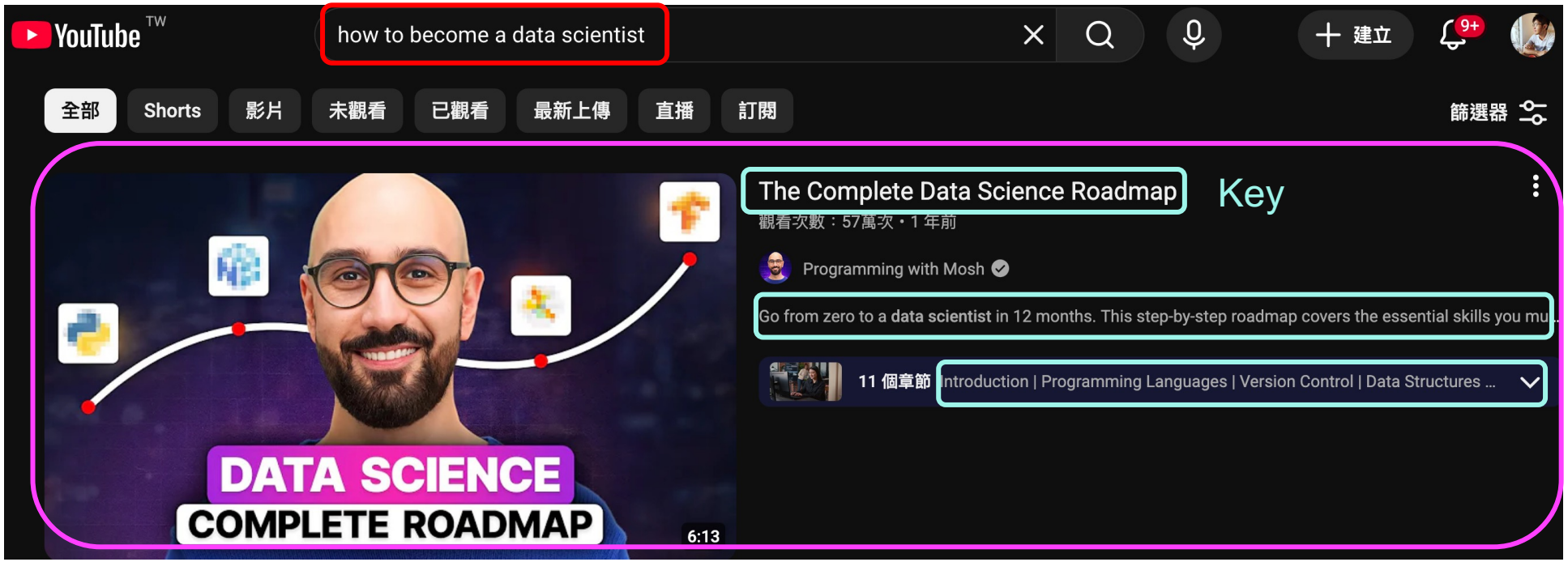
Query (Q), Key (K), and Value (V)

t = 2



QKV 的比喻

Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

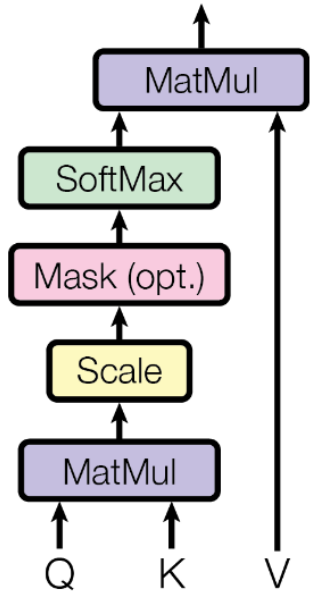


- Query: (主動) 查詢的關鍵字；Key: (被動) 查詢的對象
- Value: 值，關鍵字與對象的匹配程度

為什麼 Transformers 可以平行化？

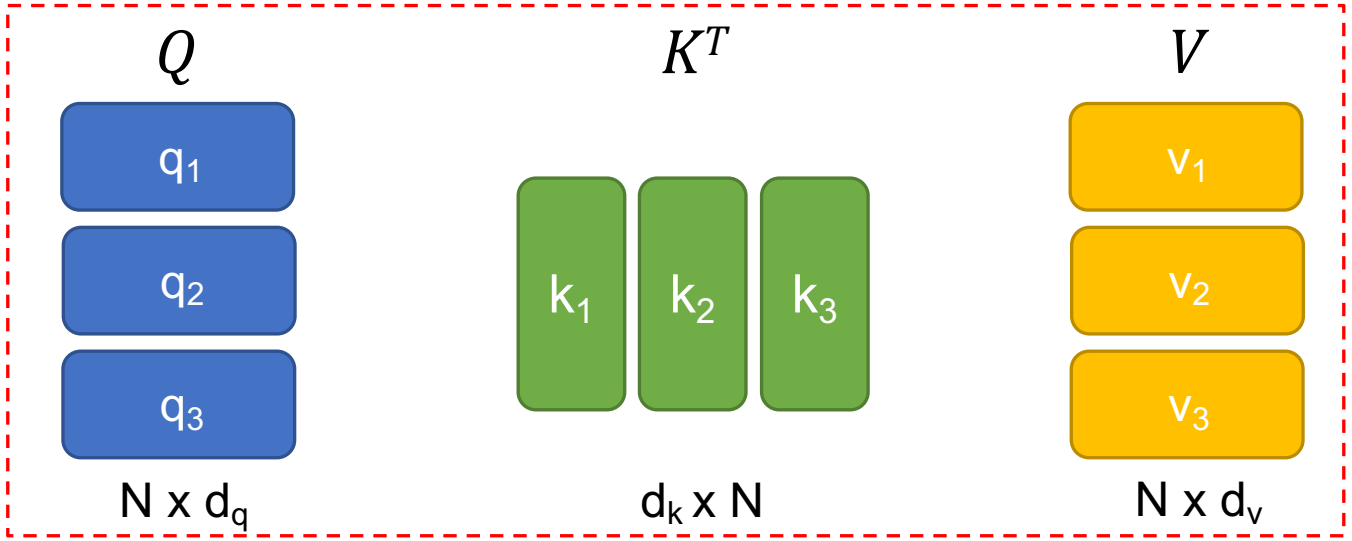
$d_q = d_k = d_v$

Scaled Dot-Product Attention



$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

(自注意力機制可以利用矩陣乘積來進行平行化計算)



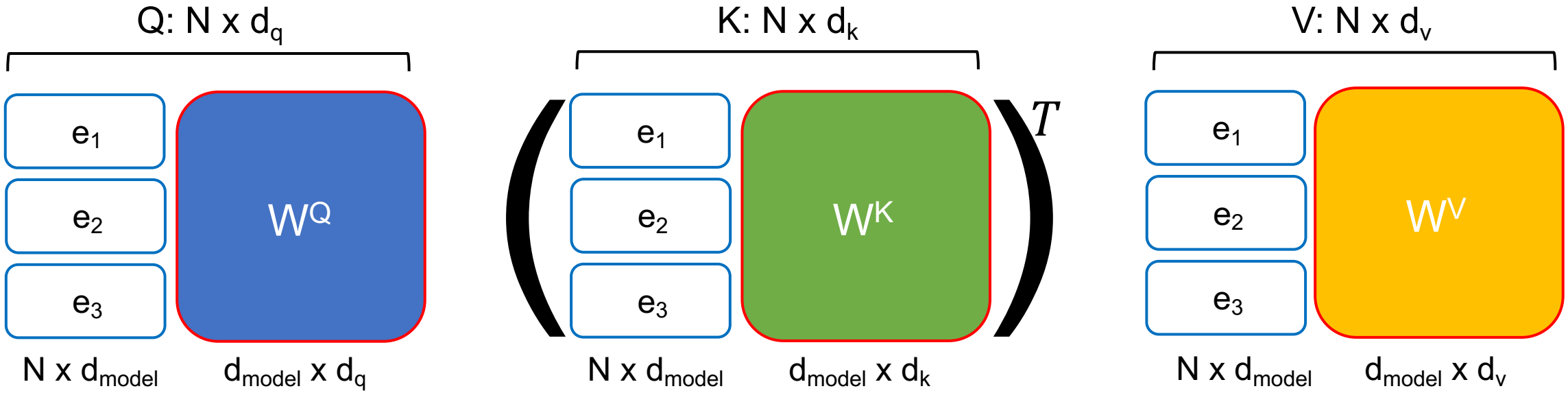
e1 I

e2 printed

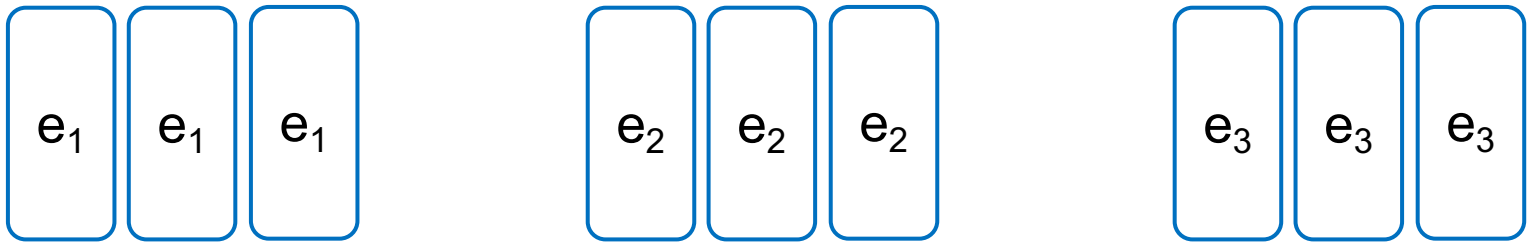
e3 Hello

Transformers 的權重值 (1/2)

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

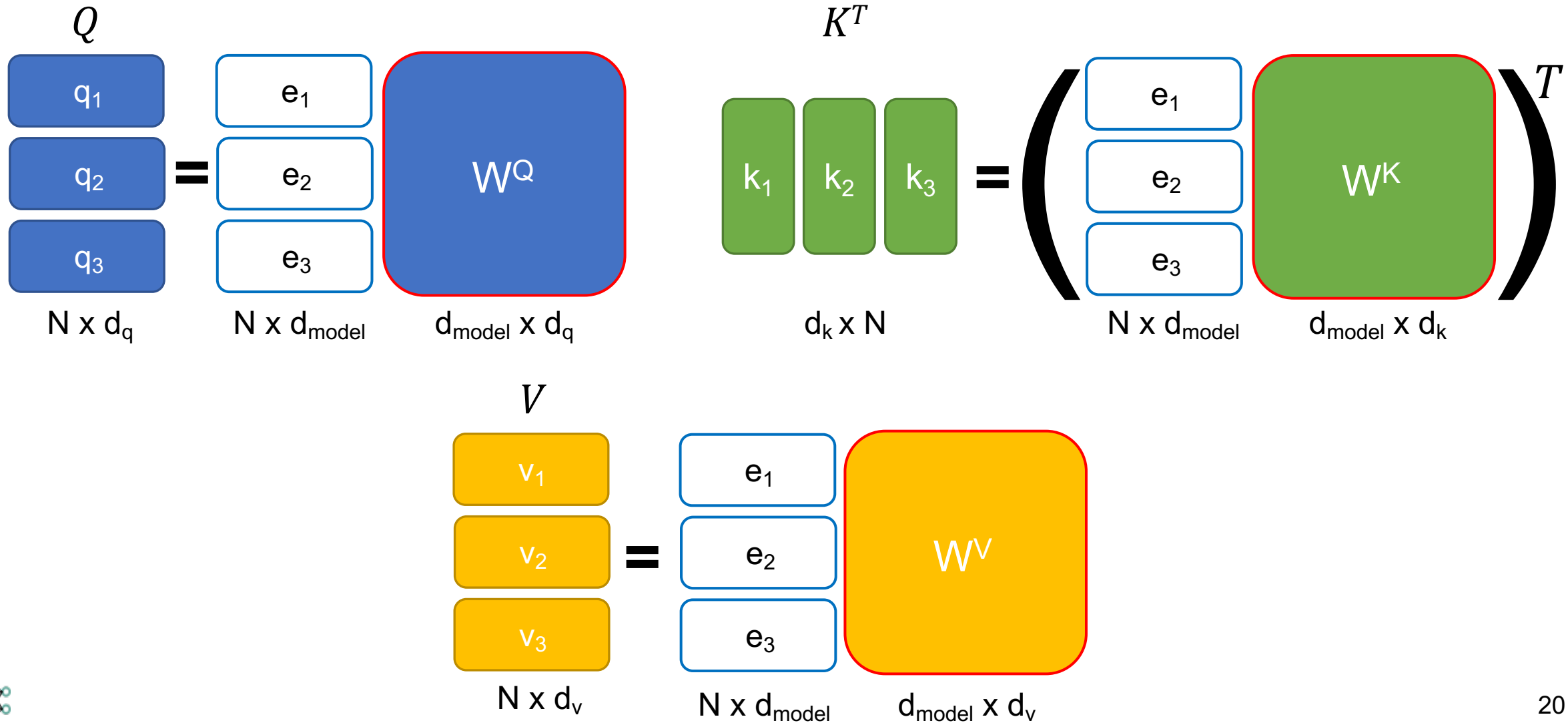


d_{model} : Embedding 的維度大小

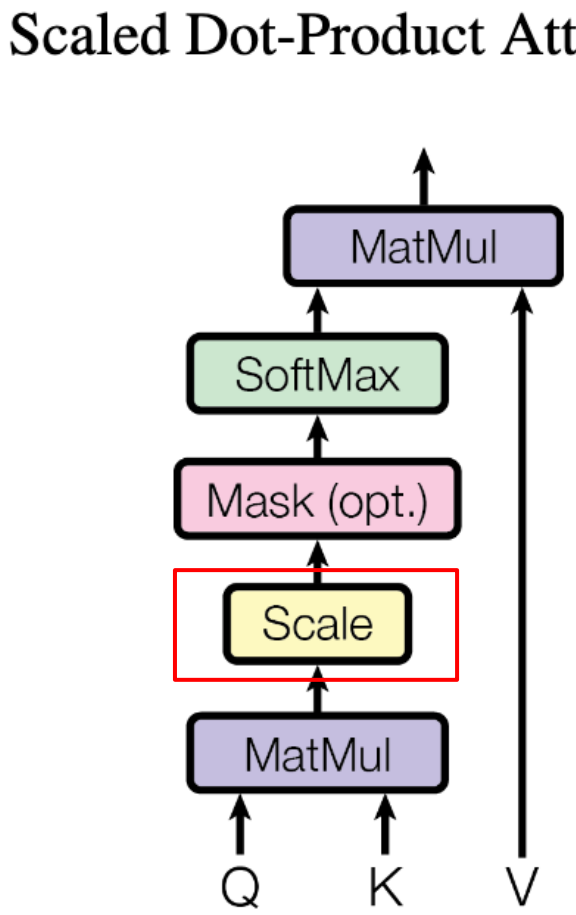


Transformers 的權重值 (2/2)

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



Self-attention 的公式與縮放項

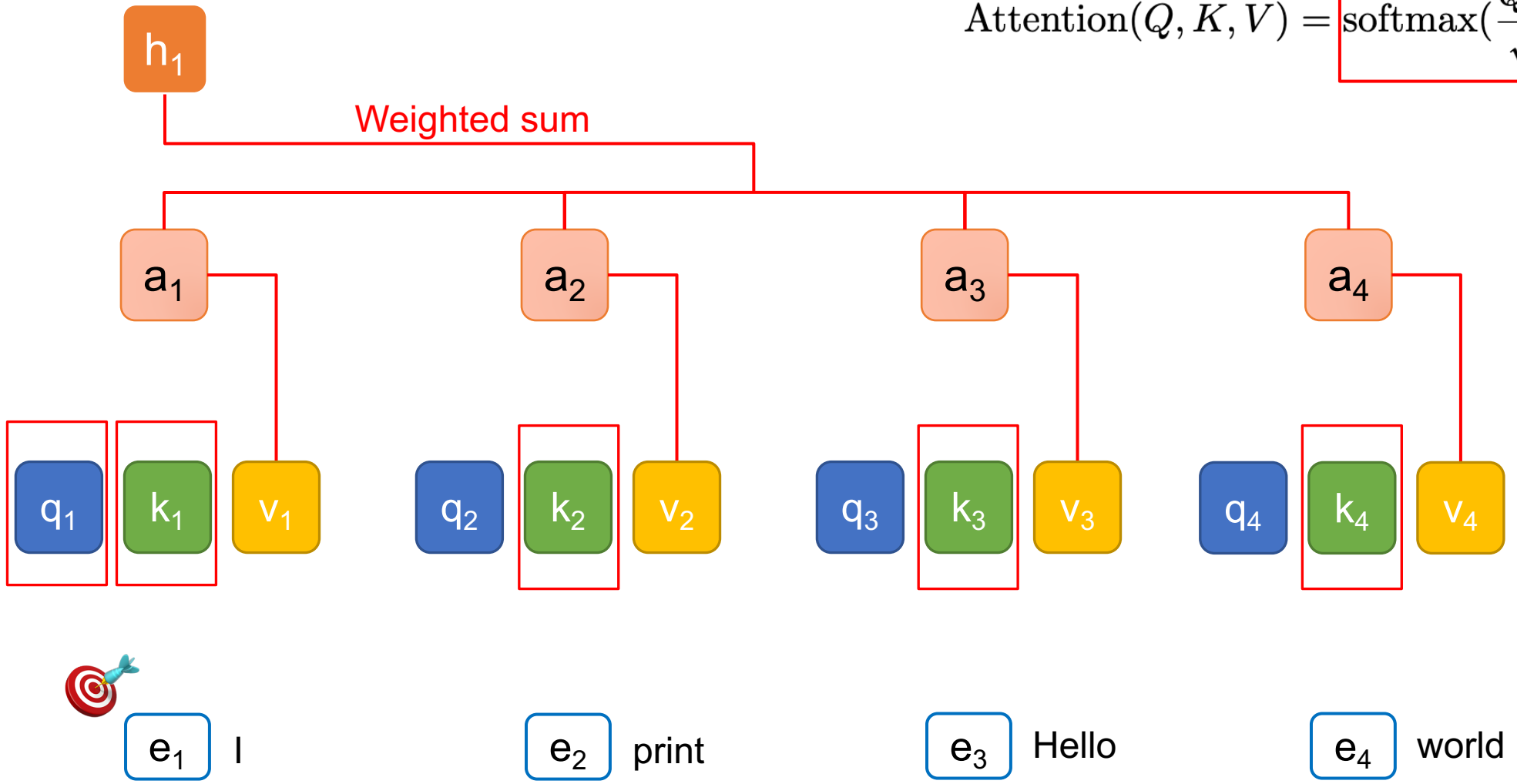


$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Query (Q), Key (K), and Value (V)

t = 1

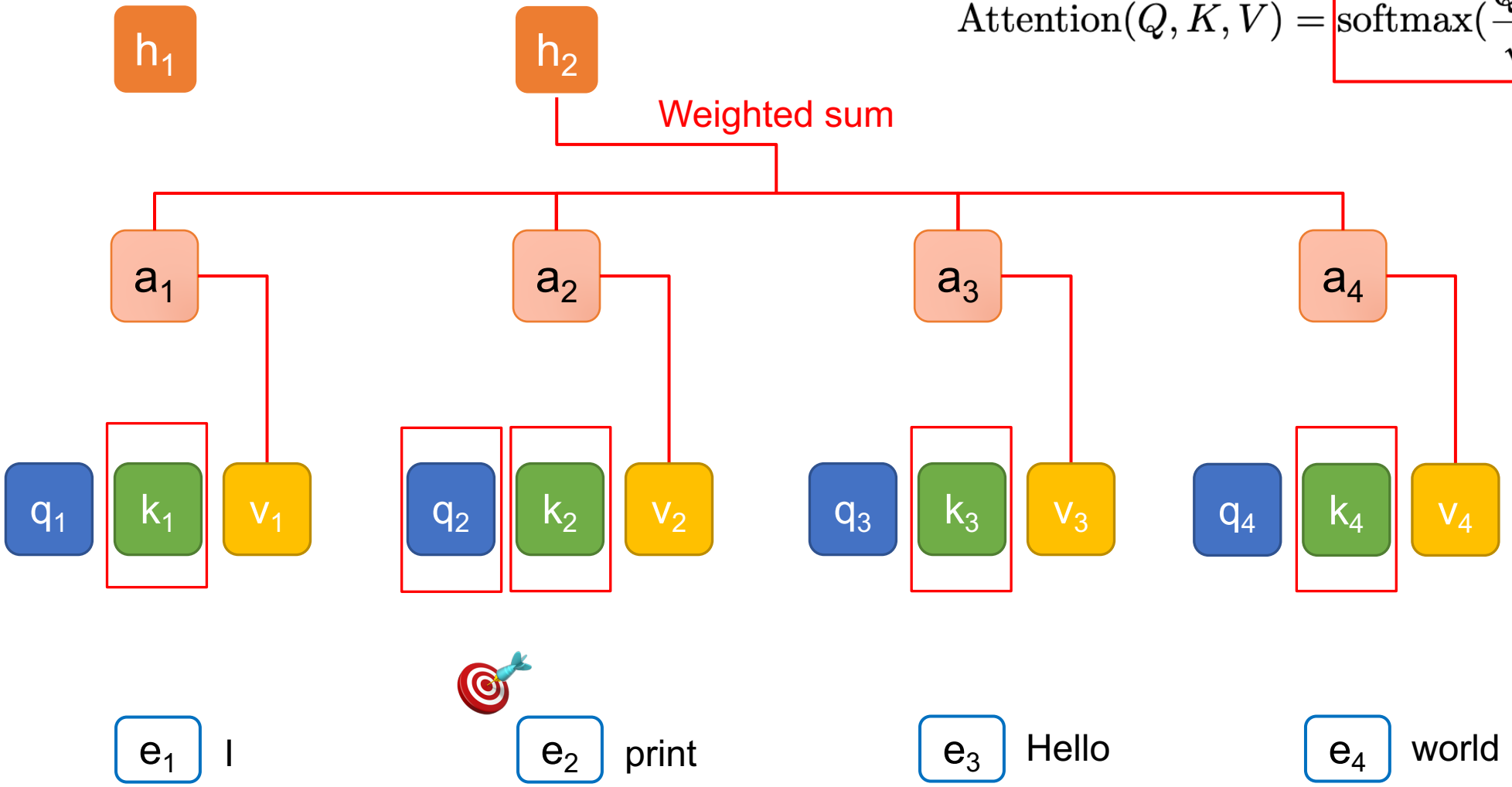
$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



Query (Q), Key (K), and Value (V)

t = 2

$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$



Self-attention 動畫

Figure source: https://research.google/blog/transformer-a-novel-neural-network-architecture-for-language-understanding/

Attention Is All You Need

Essential AI

Ashish Vaswani* Google Brain avaswani@google.com

Noam Shazeer* Google Brain noam@google.com

Anthropic

Niki Parmar* Google Research nikip@google.com

Inceptive

Jakob Uszkoreit* Google Research usz@google.com

Sakana AI

Llion Jones* Google Research llion@google.com

Cohere

Aidan N. Gomez* † University of Toronto aidan@cs.toronto.edu

OpenAI

Łukasz Kaiser* Google Brain lukaszkaiser@google.com

NEAR

Illia Polosukhin* ‡ illia.polosukhin@gmail.com

https://arxiv.org/abs/1706.03762v6

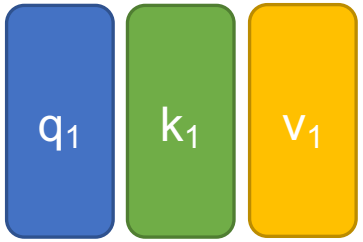
Multi-Head Attention (MHA)

- 到目前為止，我們已經介紹了 self-attention
  - We query the search engine
  - We search engine tries to map our query to the keys
  - The outputs are the attended values (e.g., the best matched videos).
- What if we want to have queries focusing on different aspects?
  - For example, one set of queries focusing on semantic similarity, another set focusing on passive/active voice.
  - **We need multiple attention mechanisms, i.e. multiple (attention) heads!**

Multi-Head Attention (MHA)

上標: head
下標: token index

Head = 1 (without MHA)

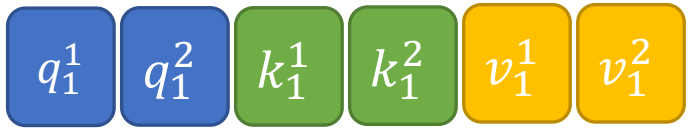


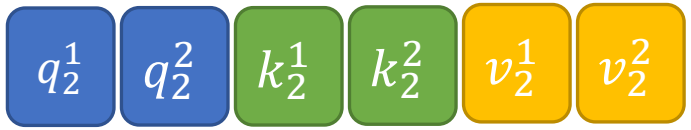
e1 I

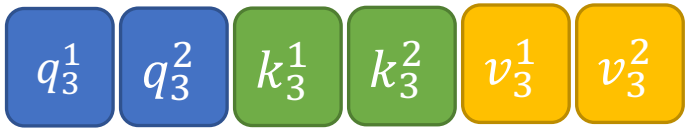
e2 printed

e3 Hello

Head = 2 (with MHA)







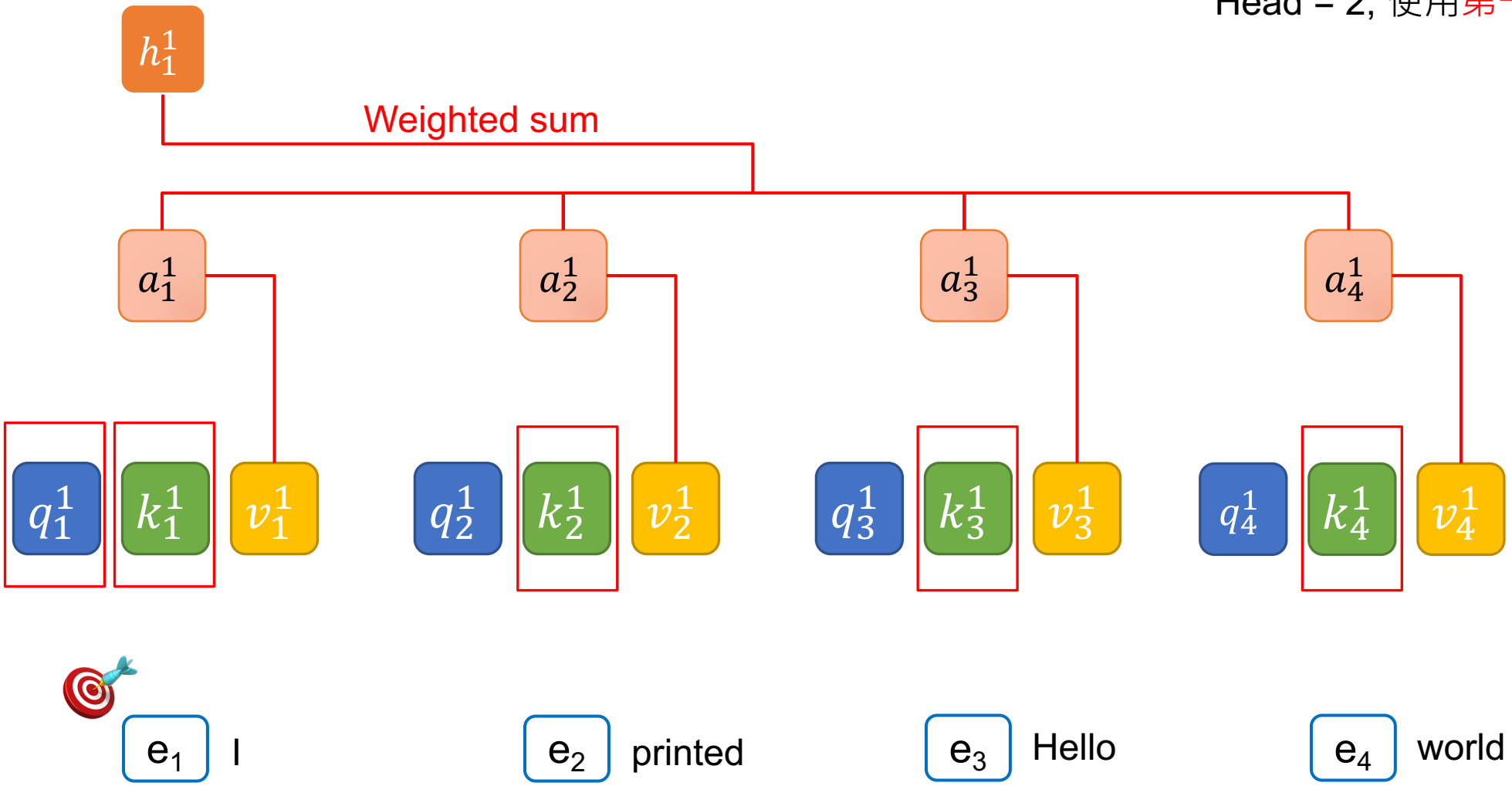
e1 I

e2 printed

e3 Hello

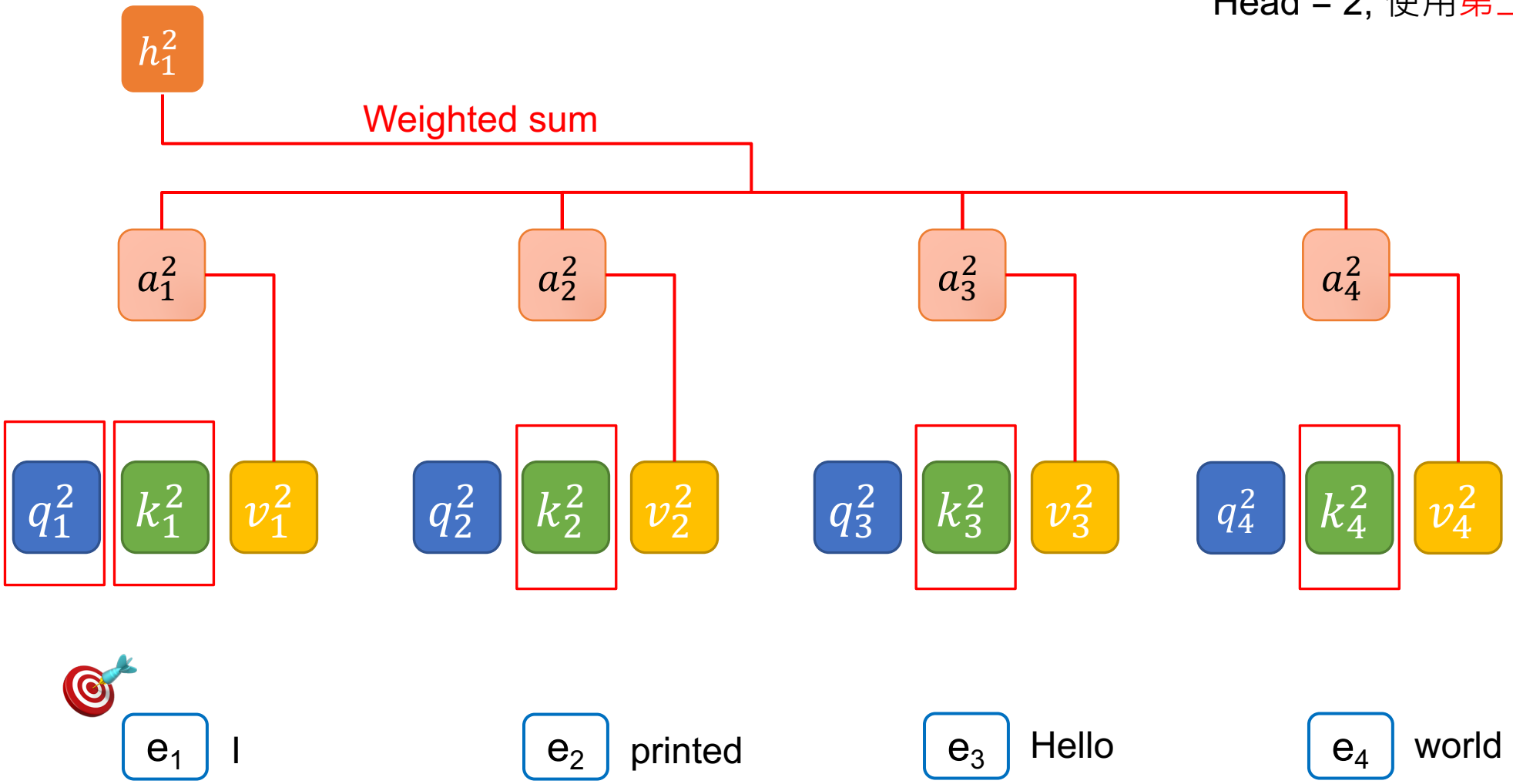
Query (Q), Key (K), and Value (V) with MHA

Head = 2, 使用第一個 head



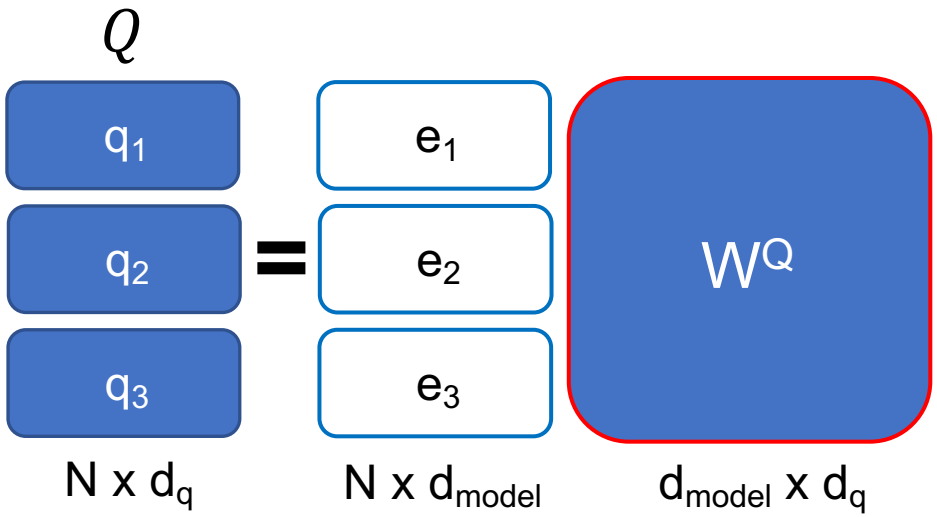
Query (Q), Key (K), and Value (V) with MHA

Head = 2, 使用第二個 head

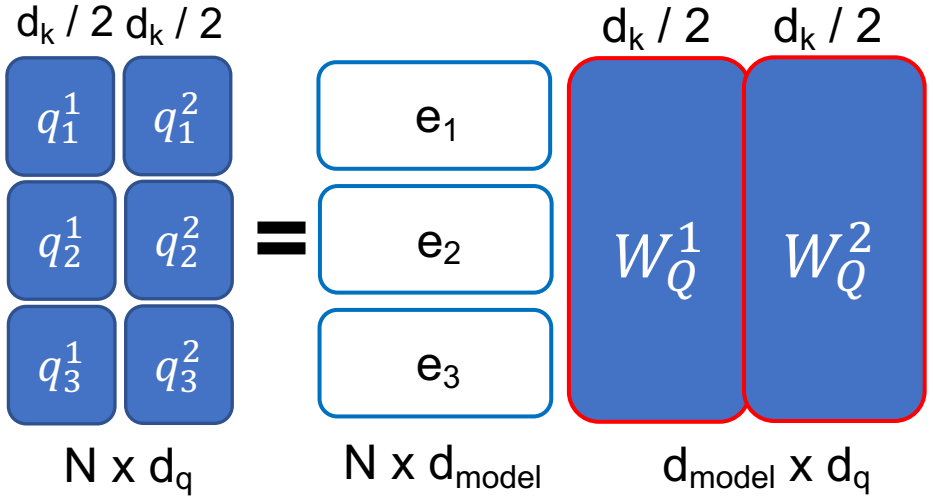


Input Dimensions of MHA

Head = 1 (without MHA)



Head = 2 (with MHA)

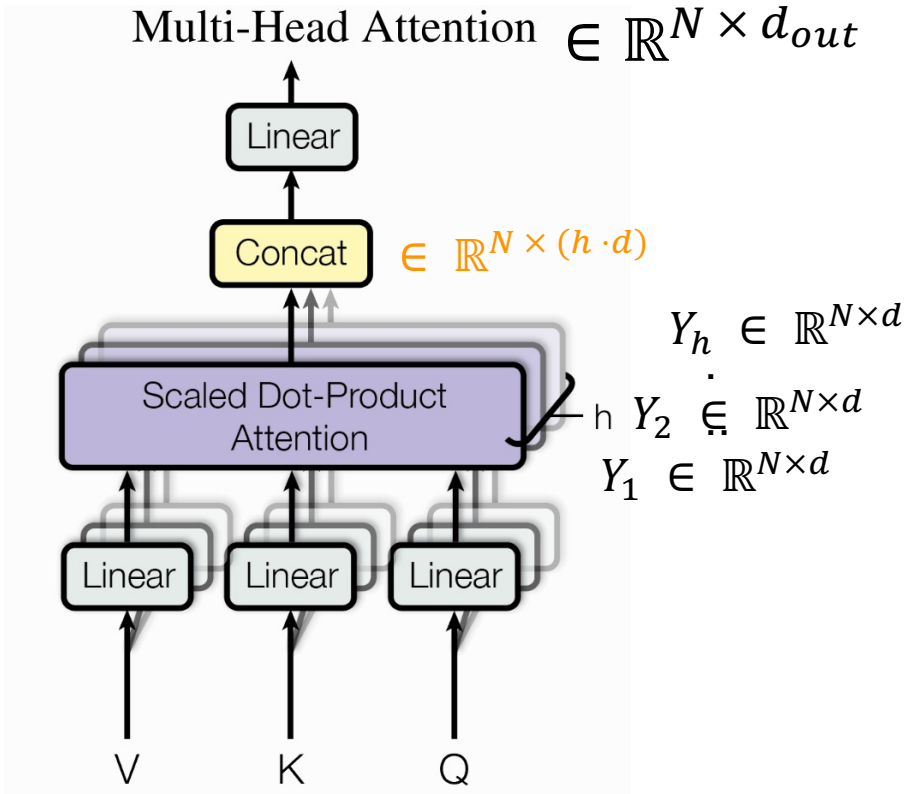


Output Dimensions

$$\text{Multihead}(Q, K, V) = \underbrace{\text{Concat}(\text{head}_1, \ldots, \text{head}_h)}_{\in\ \mathbb{R}^{N \times (h \cdot d)}} \underbrace{W^O}_{\in\ \mathbb{R}^{(h \cdot d) \times d_{out}}}$$

Head = 1 (without MHA)

Head = 2 (with MHA)



MHA 比較好嗎？

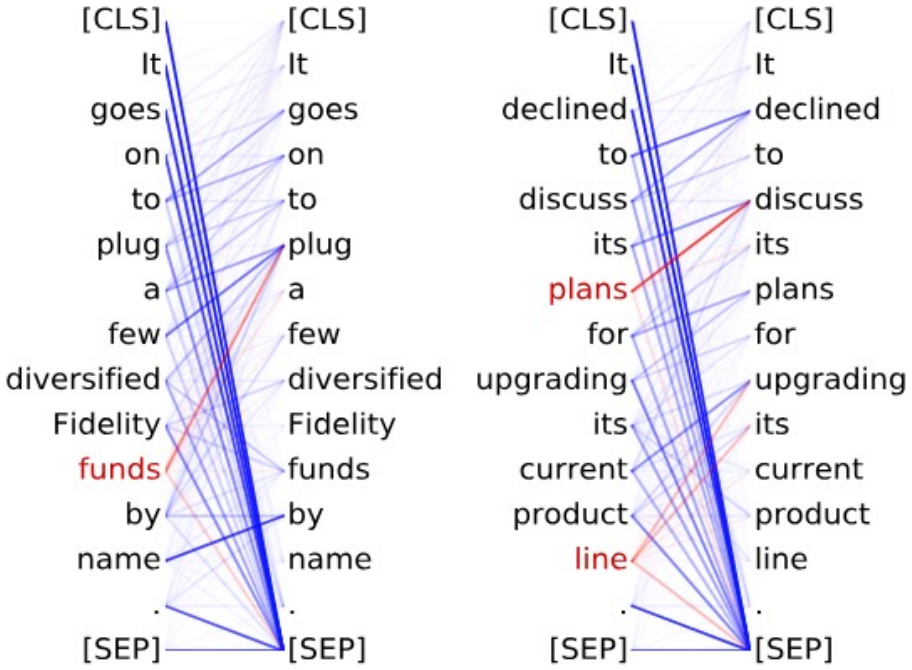
Clark, Kevin, et al. "What Does BERT Look at? An Analysis of BERT's Attention." Proceedings of the 2019 ACL Workshop BlackboxNLP: Analyzing and Interpreting Neural Networks for NLP. 2019.

實驗證明：不同 head 關注的地方不太一樣

<layer>-<head number>

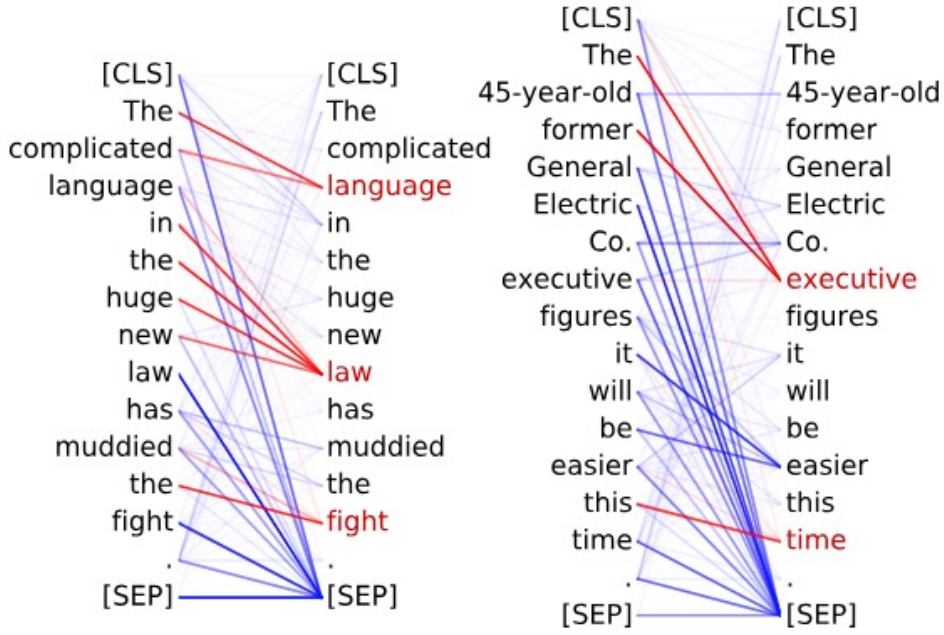
Head 8-10 關注受詞

- **Direct objects** attend to their verbs
- 86.8% accuracy at the `dobj` relation



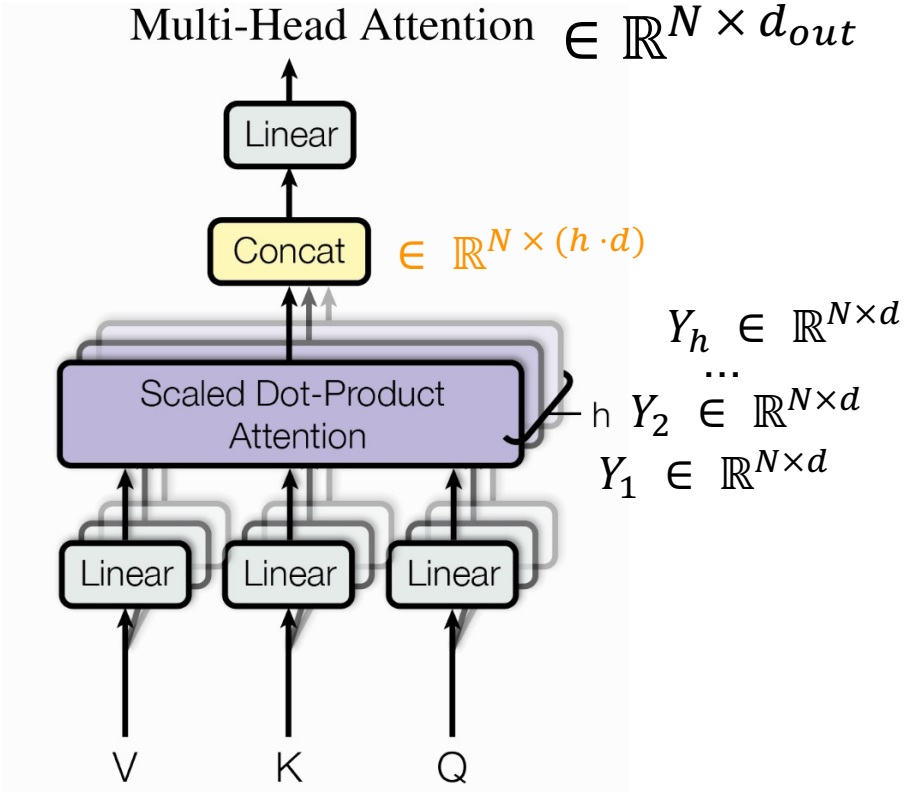
Head 8-11 關注限定詞

- **Noun modifiers** (e.g., determiners) attend to their noun
- 94.3% accuracy at the `det` relation



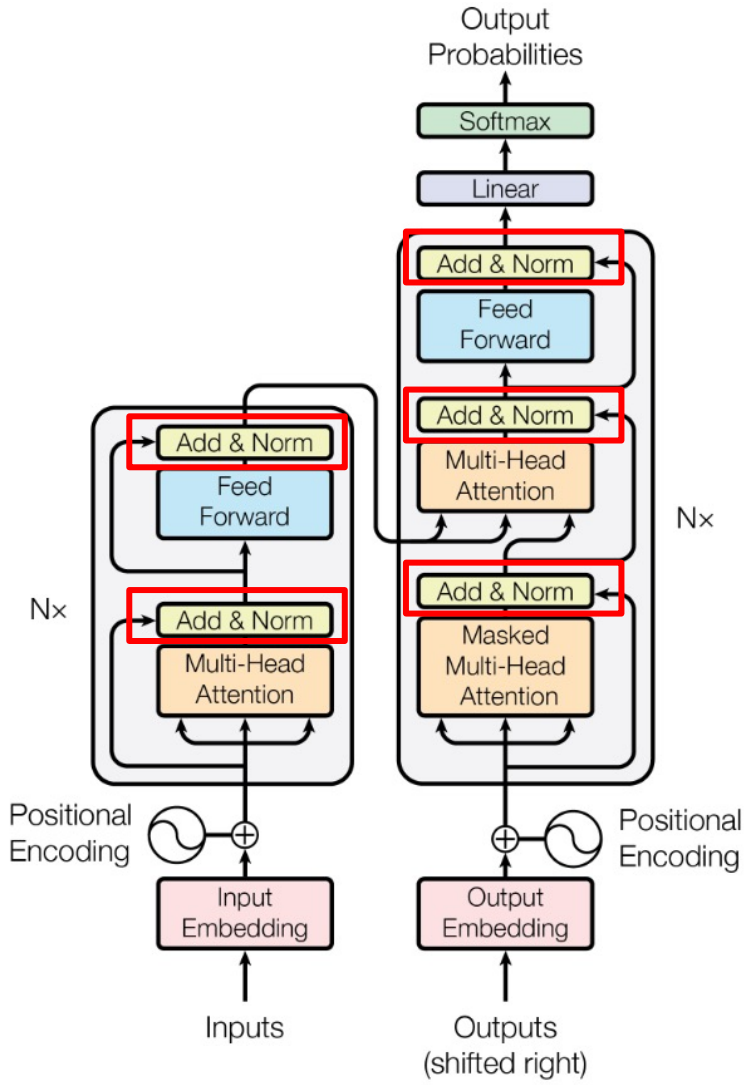
Summary of MHA

- With W^O, the concatenated head outputs can be projected to a preferred dimension (hyperparameter: d_{out}).
- No worries on how #head changes dimension of outputs!



Outline of Transformer

- Add: Residual Connection
- Norm: Layer Normalization



Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

Add & Norm

- Research shows that Residual Connection (He, Kaiming, et al., 2016) stabilizes training.
- Let the layers in between learn the residual relationship (i.e. $Y - X$).

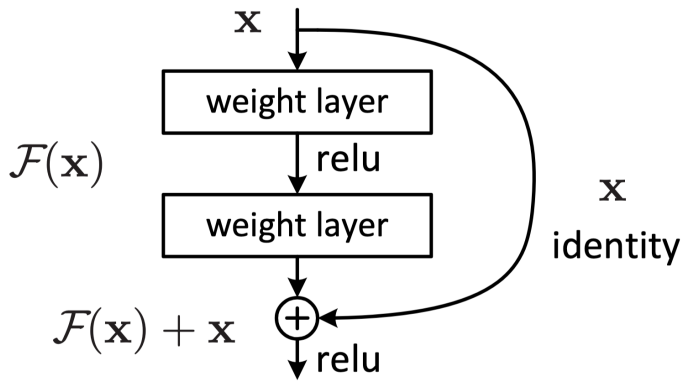
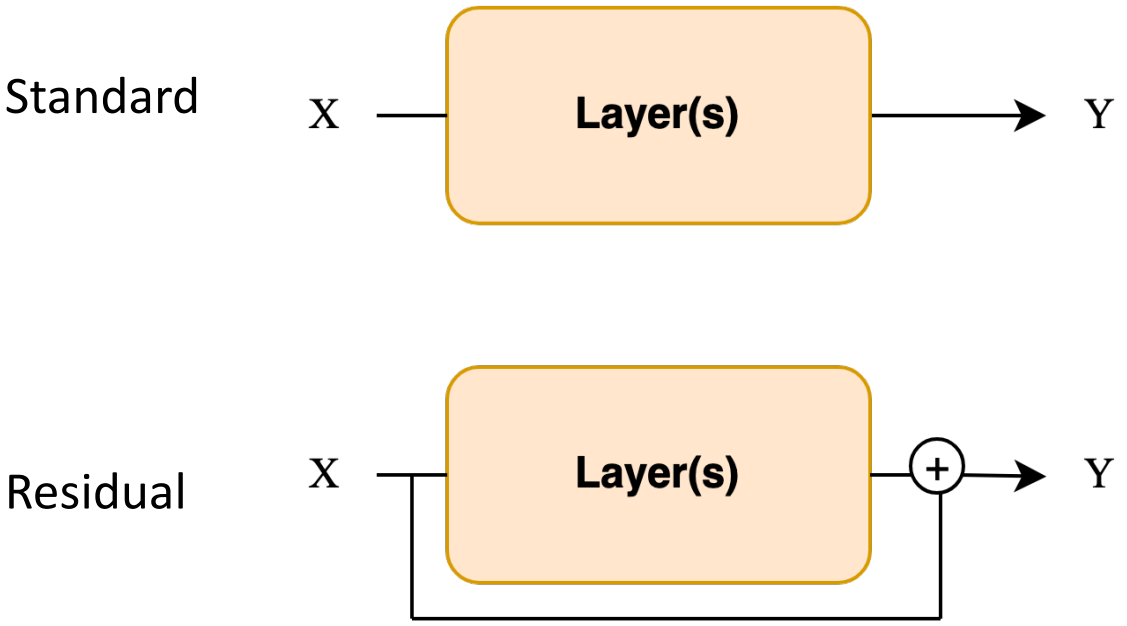


Figure 2. Residual learning: a building block.

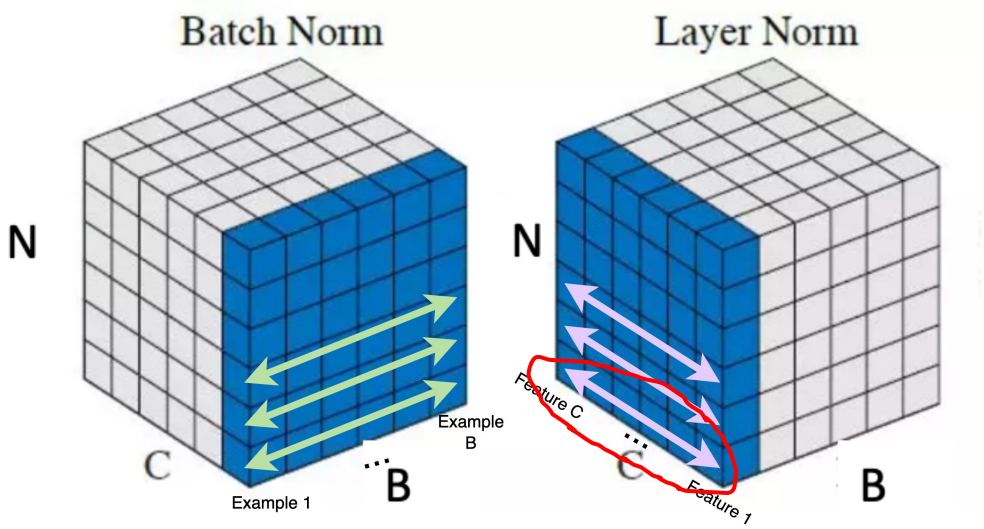


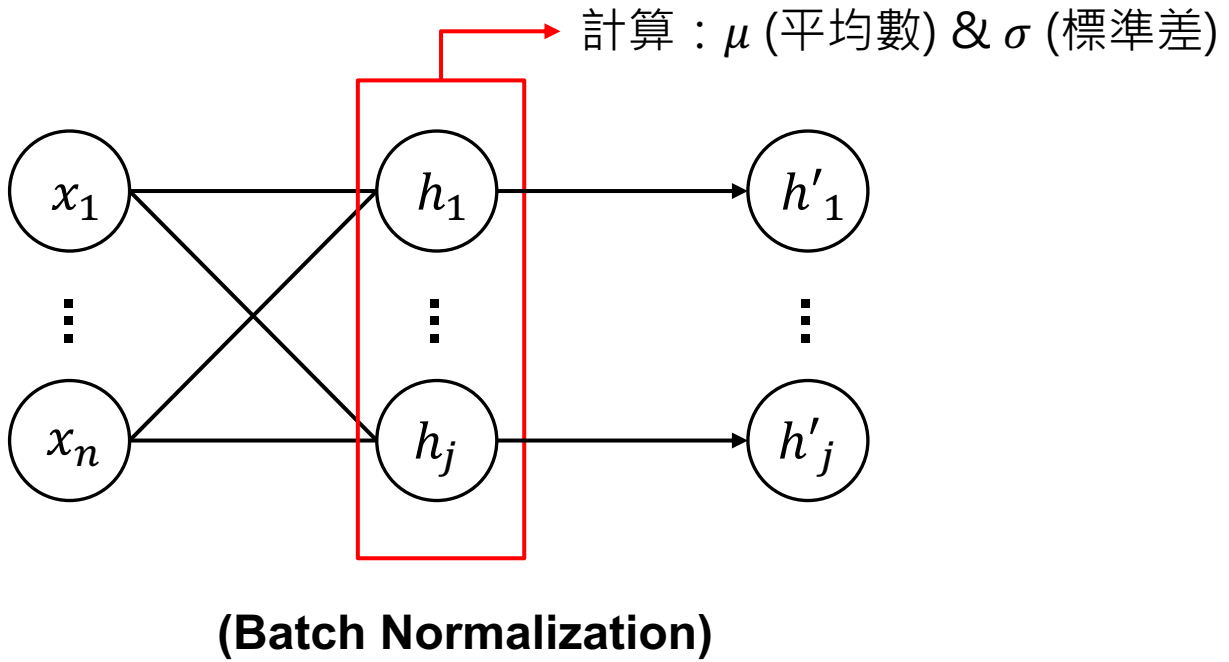
He, Kaiming, et al. "Deep residual learning for image recognition." Proceedings of the IEEE conference on computer vision and pattern recognition (2016).

Add & Norm

- Layer Norm: For an input vector, calculate its mean and std across embedding dimension (within an example).
  - $\mathbf{norm}(X) = \frac{X-\mu}{\sigma}$

B: Batch Size
C: Channels / Embedding Feature Dimension
N: Sequence length

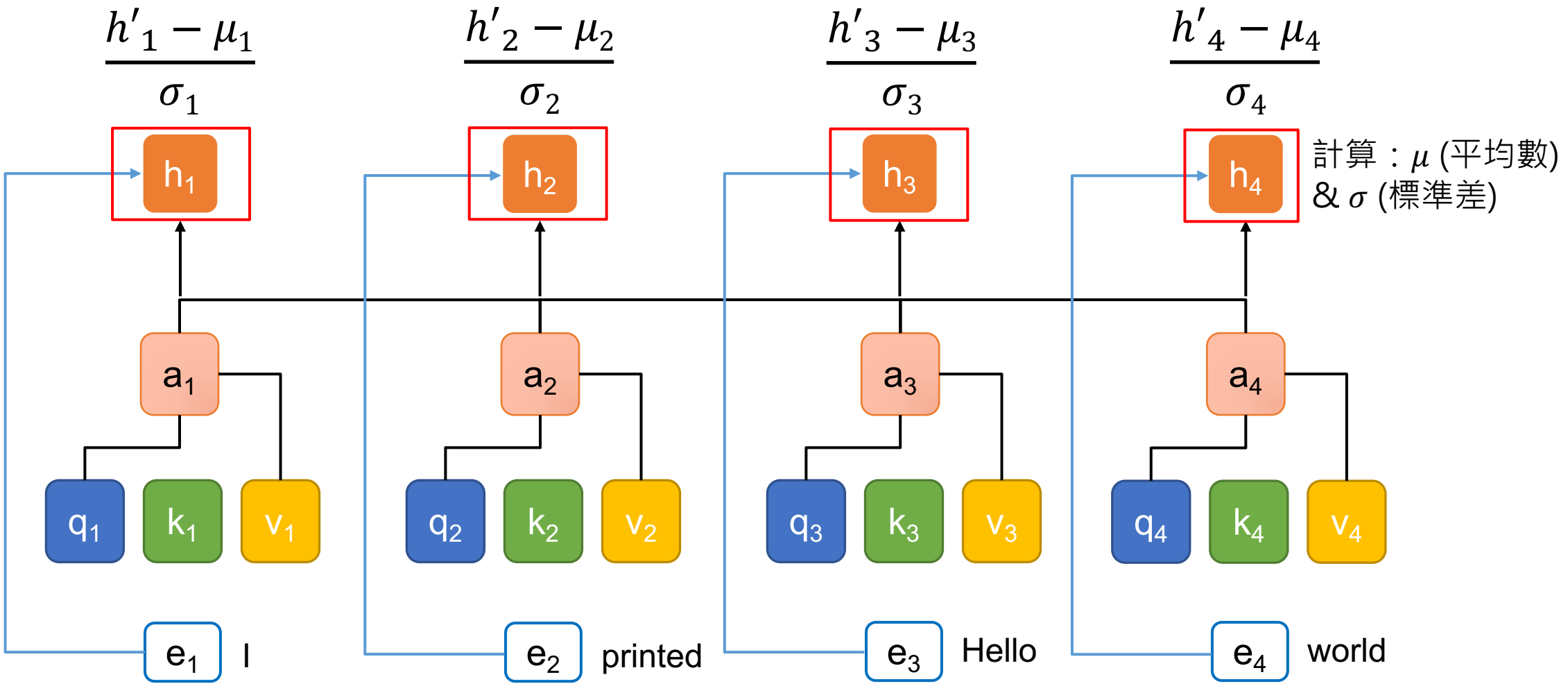




(Batch Normalization)

Ba, Jimmy Lei, Jamie Ryan Kiros, and Geoffrey E. Hinton. "Layer normalization." arXiv preprint arXiv:1607.06450 (2016).

Add & Norm



Ba, Jimmy Lei, Jamie Ryan Kiros, and Geoffrey E. Hinton. "Layer normalization." arXiv preprint arXiv:1607.06450 (2016).

Why LN not BN?

- 資料長度差距
- 模型實作表現差異

[Intro] Static padding

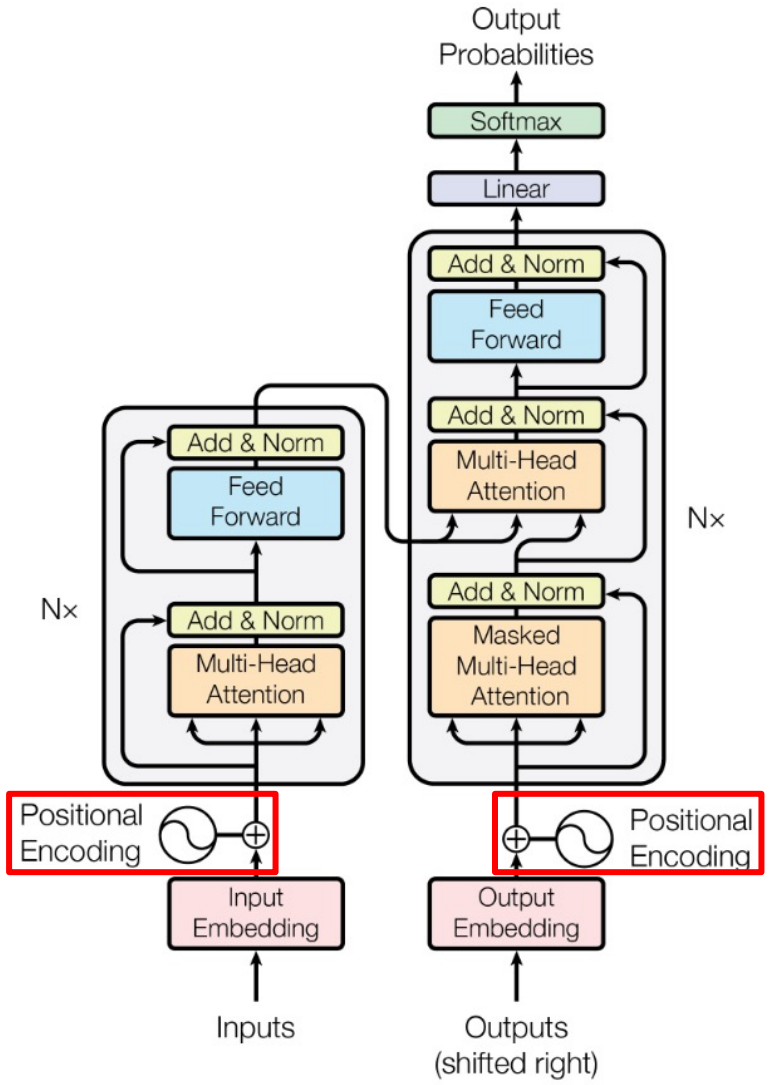
batch 1

I	love	CGU	[PAD]	[PAD]	[PAD]	
Yesterday	I	watched	a	good	movie	longest
Your	new	shoes	look	nice	[PAD]	

batch 2

Merry	Christmas	[PAD]	[PAD]	[PAD]	[PAD]
I	love	coding	[PAD]	[PAD]	[PAD]
Me	[PAD]	[PAD]	[PAD]	[PAD]	[PAD]

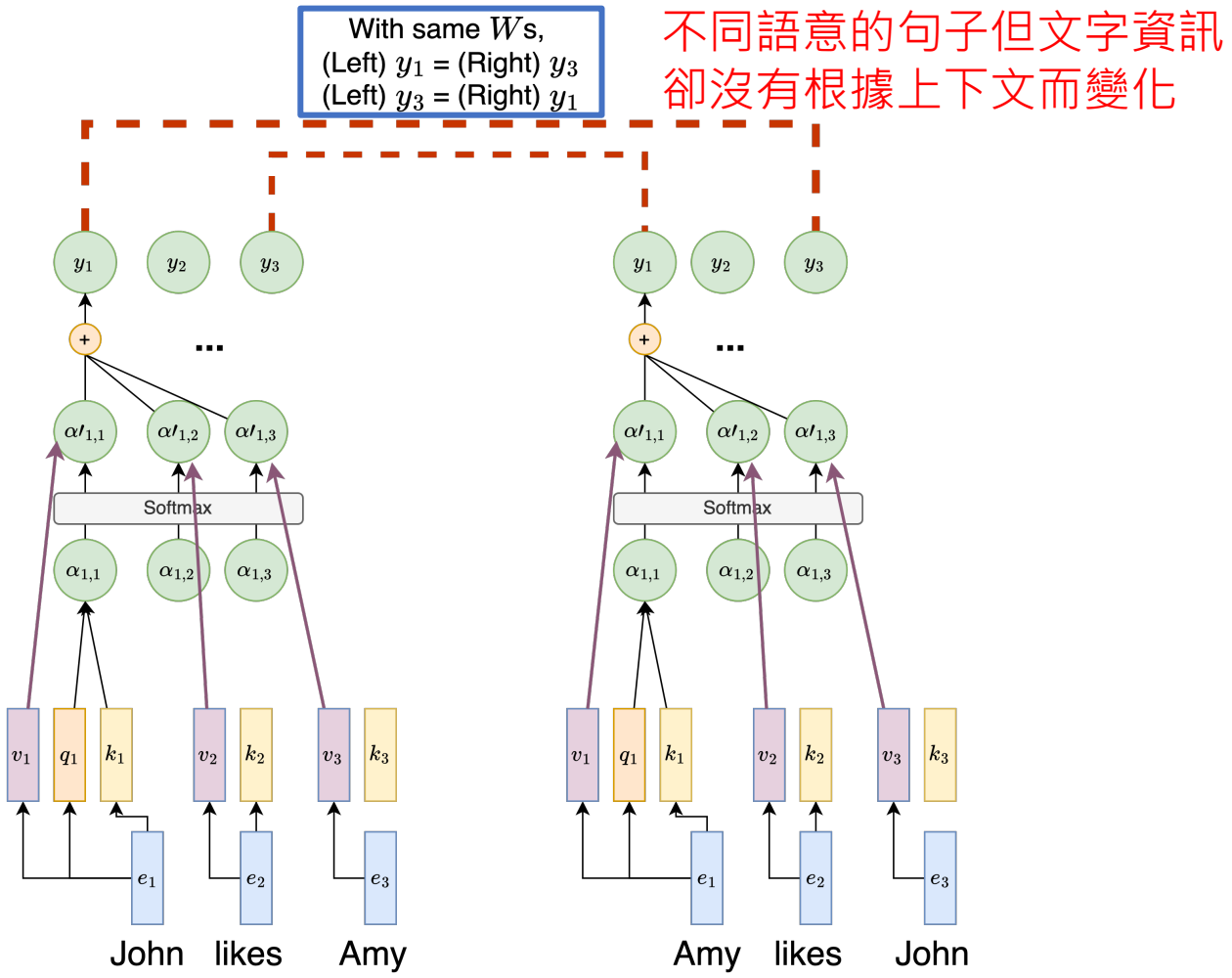
Outline of Transformer



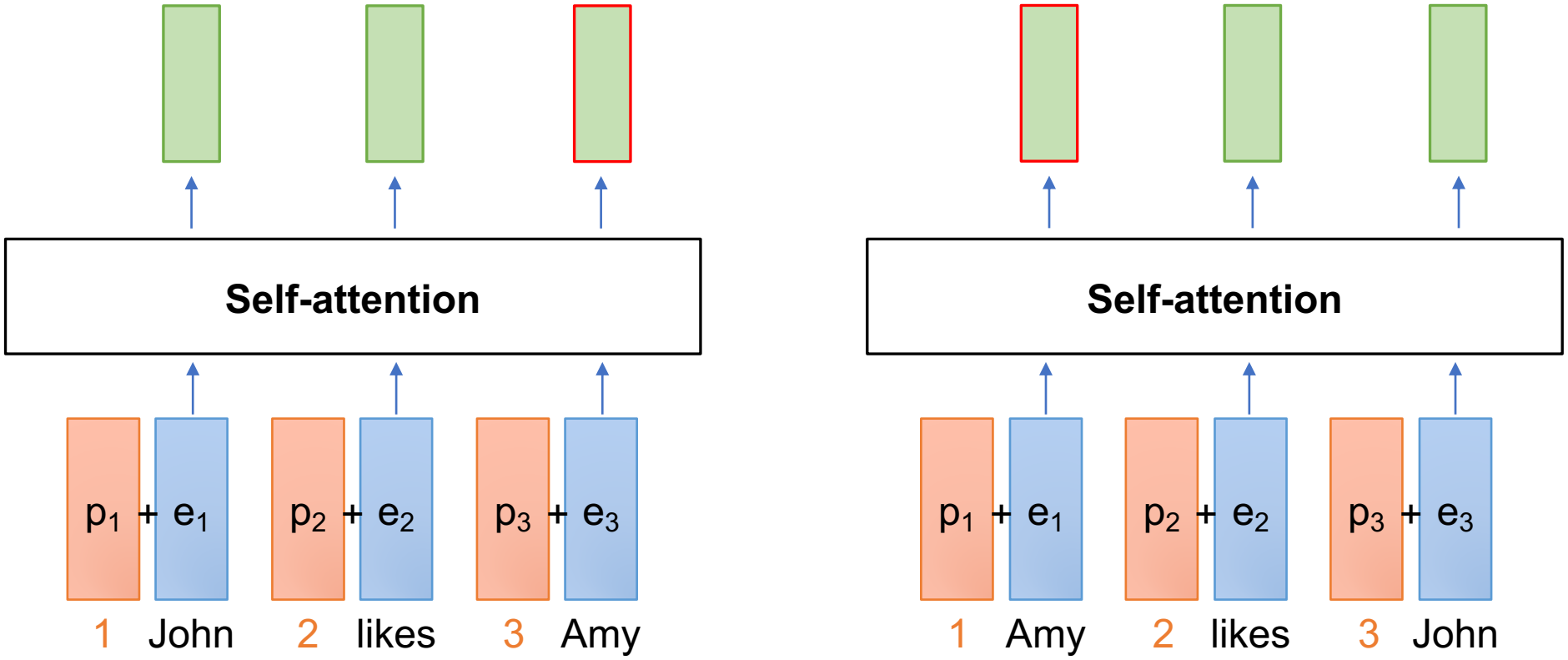
Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

Lack of Position Modeling

- RNN 有位置資訊 (因為 RNN step-by-step 處理文字)
- 然而，Transformer 中 self-attention 的操作遺失了 step-by-step 處理文字的方式，故缺少位置資訊



常見位置資訊作法：Learned Positional Embeddings



此做法又稱絕對位置資訊編碼 (Absolute positional Encoding)

Learned Positional Embeddings 程式碼

https://huggingface.co/transformers/v2.1.1/_modules/transformers/modeling_bert.html

```
class BertEmbeddings(nn.Module):
    """Construct the embeddings from word, position and token_type embeddings.
    """
    def __init__(self, config):
        super(BertEmbeddings, self).__init__()
        self.word_embeddings = nn.Embedding(config.vocab_size, config.hidden_size, padding_idx=0)
        self.position_embeddings = nn.Embedding(config.max_position_embeddings, config.hidden_size)
        self.token_type_embeddings = nn.Embedding(config.type_vocab_size, config.hidden_size)

        # self.LayerNorm is not snake-cased to stick with TensorFlow model variable name and be able to load
        # any TensorFlow checkpoint file
        self.LayerNorm = BertLayerNorm(config.hidden_size, eps=config.layer_norm_eps)
        self.dropout = nn.Dropout(config.hidden_dropout_prob)
```

缺點：Learned Positional Embeddings 不會超出 max_position 之後的資訊

Positional Encoding in Transformer

- 創造出相對位置的 embeddings
- 核心精神：利用三角函數的週期性來建構序列資訊
  - 偶數的 embedding 單元 (2i , i = 0, 2, 4, …, d_{model} / 2 - 2)：Sin 函數
  - 奇數的 embedding 單元 (2i+1, i = 1, 3, 5, …, d_{model} / 2 - 1)： Cosine 函數

$$PE_{(pos,2i)} = sin(pos/10000^{2i/d_{\text{model}}})$$

$$PE_{(pos,2i+1)} = cos(pos/10000^{2i/d_{\text{model}}})$$

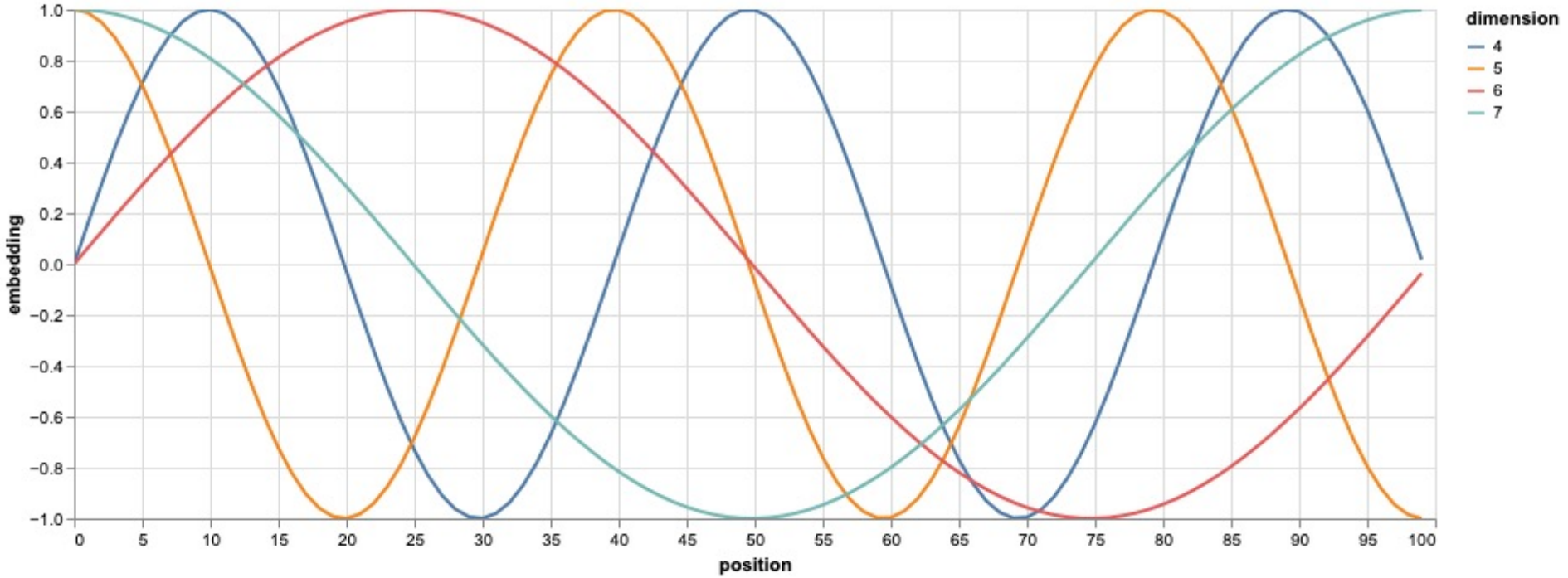
i越大代表分母的指數越大，sin的角度越小

10000是實驗得出的較佳數值

輸入序列中任一token的位置

Figure source: https://nlp.seas.harvard.edu/annotated-transformer/#positional-encoding

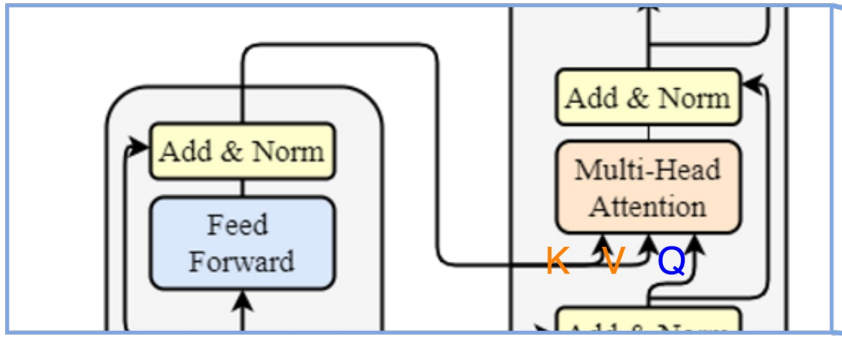
Positional Encoding Visualization



- Question: 為什麼需要同時有 Sin 和 Cosine 的函數？
- Ans: 藉由不同週期性來捕捉序列中各個位置的關係

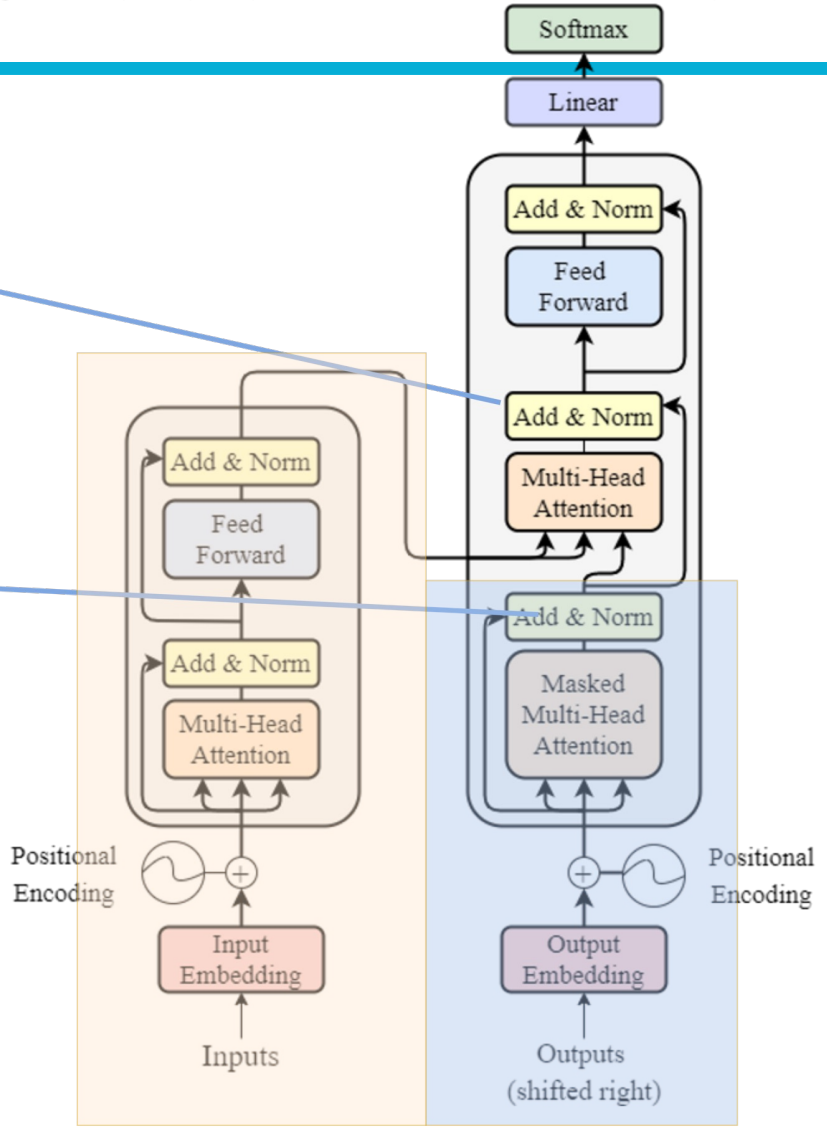
Figure source: https://nlp.seas.harvard.edu/annotated-transformer/#positional-encoding

Encoder-Decoder 橋樑: Cross-Attention



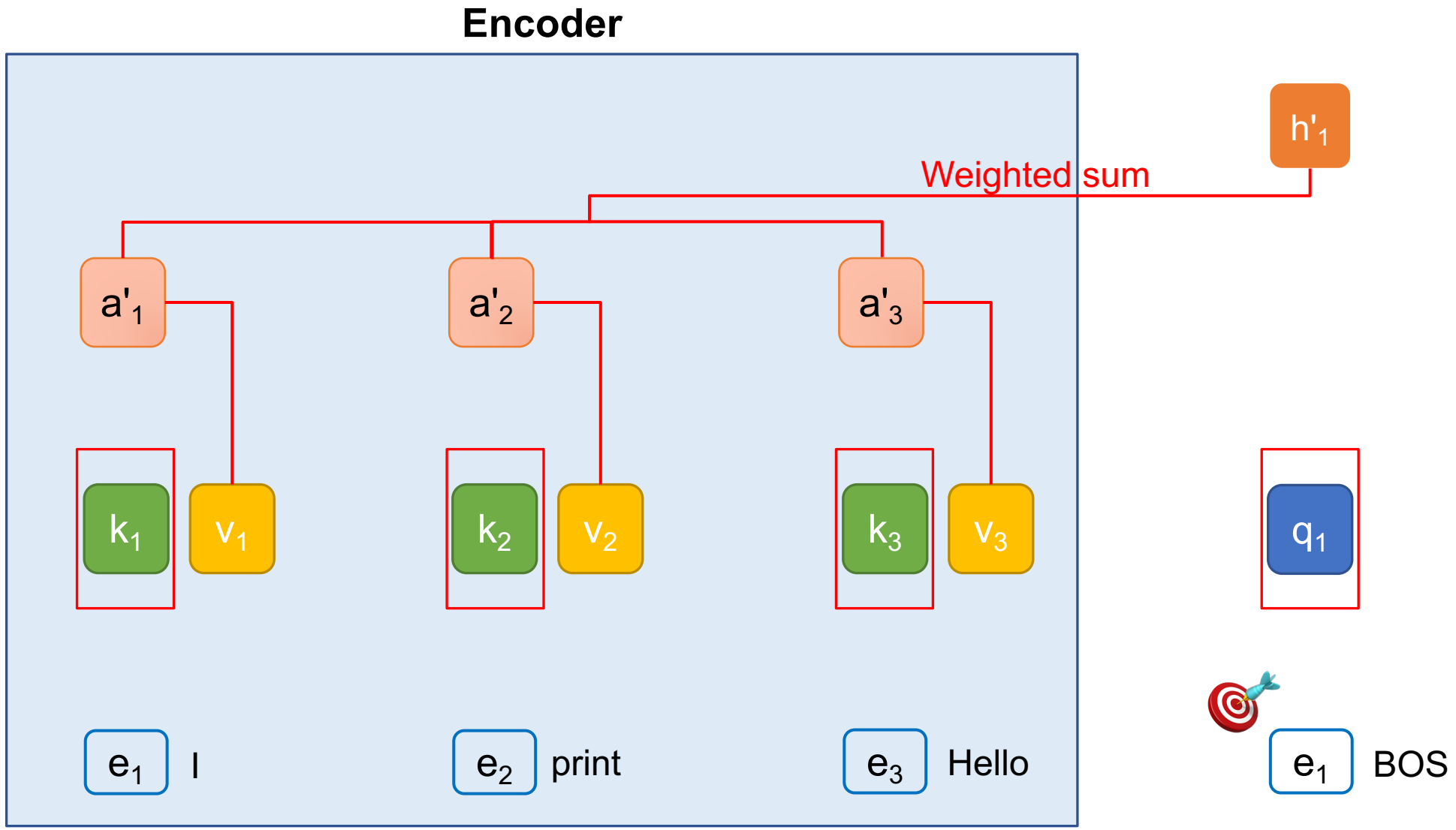
K, V uses the output from encoder as their X;

Q uses the decoder's information as X.



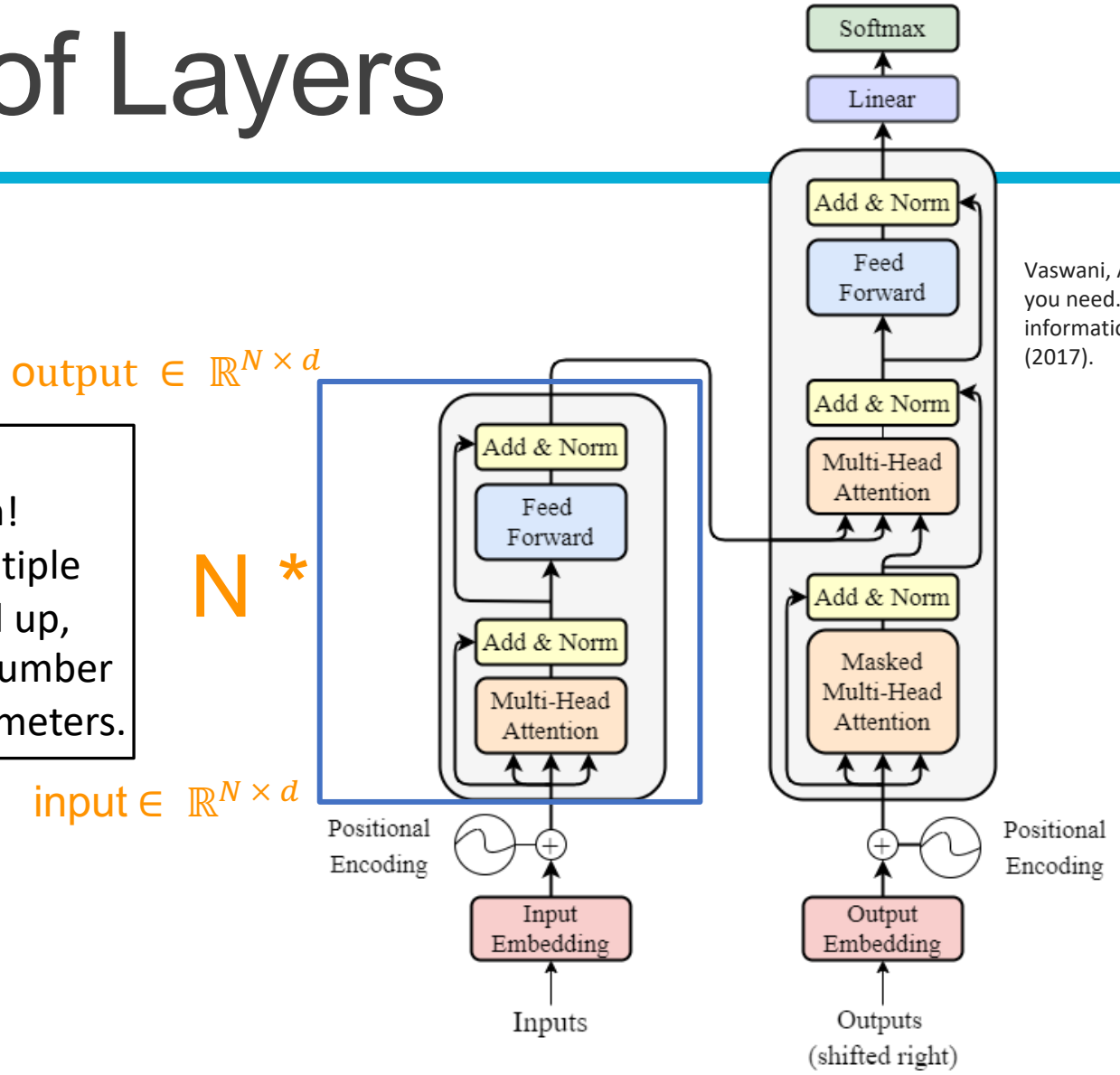
$$\text{Attention}(Q, K, V) = \text{softmax}(\frac{QK^T}{\sqrt{d_k}})V$$

Cross-attention



Number of Layers

Input, output dimension match!
We can have multiple encoders stacked up, to increase the number of trainable parameters.

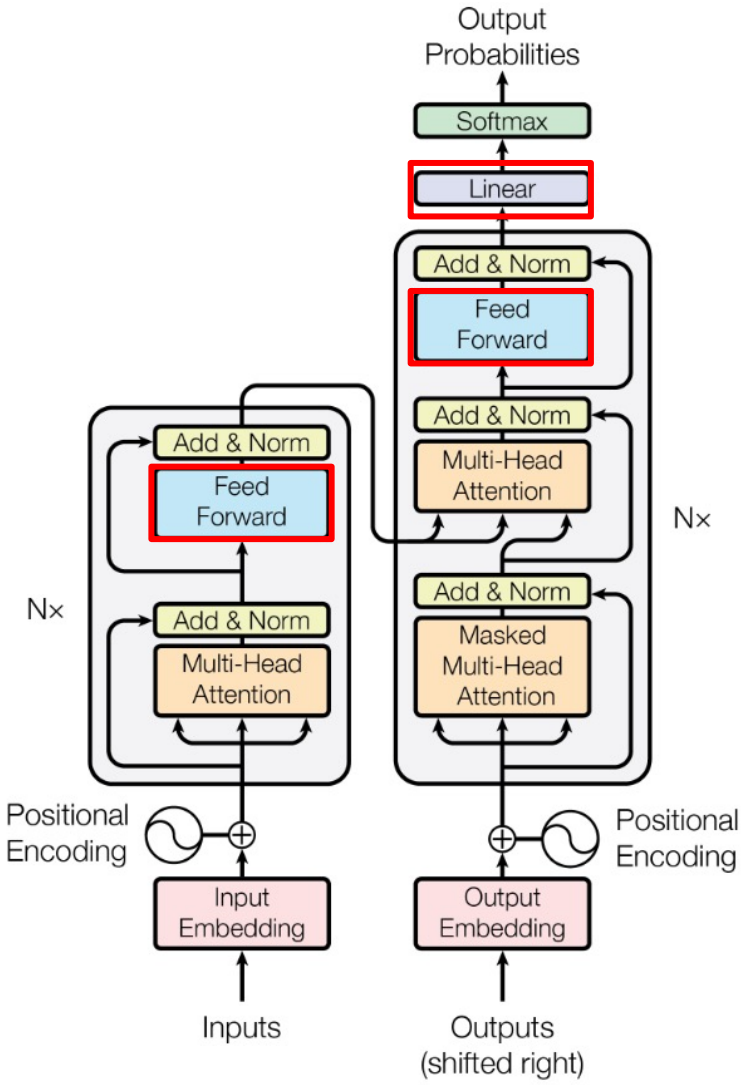


Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).

Outline of Transformer

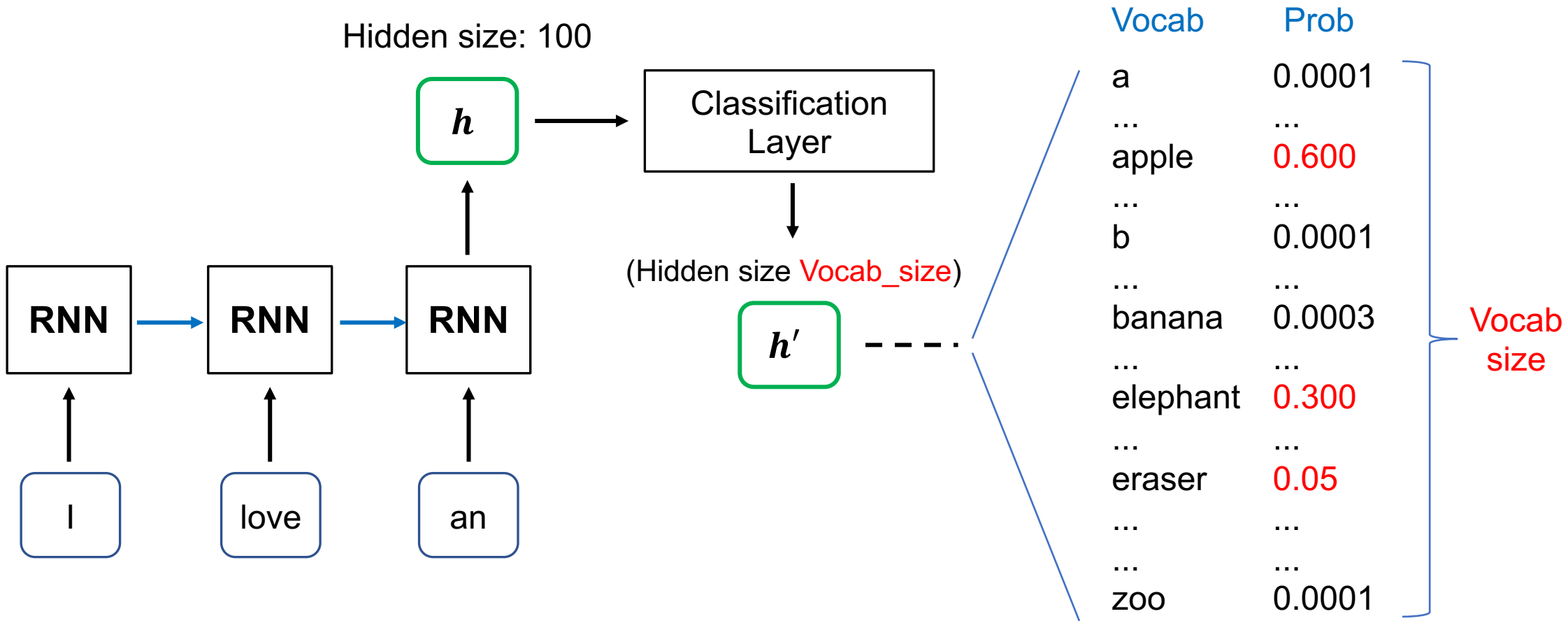
Feed Forward Networks:

- Simply multiply by a weight matrix (MLP).
- Used to project to a specific dimension.



Vaswani, Ashish, et al. "Attention is all you need." NeurIPS (2017).

[Recap] Text Generation with RNN



Transformers' Achievements

1. In original paper, State-of-The-Art (SoTA) of machine translation.

Table 2: The Transformer achieves better BLEU scores than previous state-of-the-art models on the English-to-German and English-to-French newstest2014 tests at a fraction of the training cost.

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	**41.29**	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	**$3.3 \cdot 10^{18}$**	
Transformer (big)	**28.4**	**41.8**	$2.3 \cdot 10^{19}$	

6 encoders,
6 decoders,
h = 512, 100K train steps.

6 encoders,
6 decoders,
h = 1024, 300K train steps.

Vaswani, Ashish, et al. "Attention is all you need." Advances in neural information processing systems 30 (2017).
https://arxiv.org/abs/1706.03762

Transformer Variants (there are many more)

- Universal Transformer
  - Adds Adaptive Computation Time
  - Dehghani, Mostafa, et al. "Universal transformers." arXiv preprint arXiv:1807.03819 (2018).
- Longformer
  - Tackles long documents
  - Iz Beltagy, et al. (2020). Longformer: The Long-Document Transformer. arXiv:2004.05150.
- Roformer
  - Changes the design of positional encoding
  - Su, Jianlin, et al. "RoFormer: Enhanced Transformer with Rotary Position Embedding. CoRR abs/2104.09864 (2021)." arXiv preprint arXiv:2104.09864 (2021).
- Vision Transformer
  - Tackles vision tasks
  - Alexey Dosovitskiy, et al. “An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.” International Conference on Learning Representations. 2021.

Tokens vs. words

- token 是 (語言) 模型在每個時間點處理的單位
- word 是語言本身的單位
  - token 可以是 word，也可以是 sub-word
  - 一個 word 可以是一個 token，但單純講 token 不一定指的是 word

Traditional word tokenization

Sub-word Tokenization

Transformer 採用此方式

Transformer 架構常用於現代 AI 模型

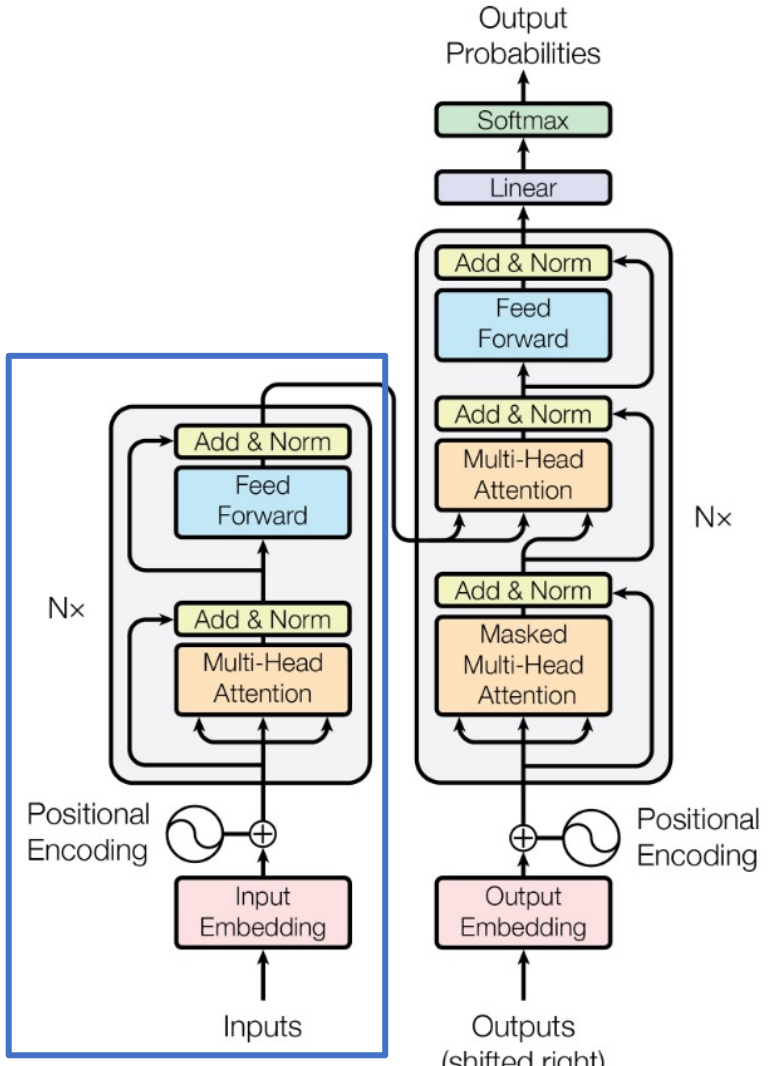
For example:

- BERT (Bidirectional encoder representations from transformers)
  - Devlin et al., 2018
- GPT series

(BERT)

N=12

Encoder



Additional Links

- https://nlp.seas.harvard.edu/annotated-transformer/
- https://datascience.stackexchange.com/questions/82451/why-is-10000-used-as-the-denominator-in-positional-encodings-in-the-transformer

Announcements

Project checkpoints

- ~~Week 9: 確定各組的題目~~
- Week 12: 進度報告 PPT (5 pages)
- Week 14: 進度報告 PPT (5+5 pages), Presentations (selected teams)
- Week 16: Final presentations for all teams
- Week 16-17: 繳交書面報告以及程式碼

Project 各個階段分數佔比

Final Project 佔學期總成績 40%

查核點 (週次)	對象: 繳交內容	分數佔比
Checkpoint1 (Week 12)	All teams: 進度報告 PPT (5 pages)檔案	5%
Checkpoint2 (Week 14)	All teams: 進度報告 PPT (5+5 pages*)檔案 Selected teams: 取6組 (1題目2組) 於課堂中報告，1組 10min	5%
Checkpoint3 (Week 16)	All teams: 最終口頭報告	15%
Checkpoint4 (Week 16-17)	All teams: 書面報告檔案 (含程式碼)、小平台 (optional)	15%

*繼承Checkpoint1內容+實作

Thank you!

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